Stress and Turnover in Helping Professions: A Randomized Controlled Longitudinal Study of Head Start Employees

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This article reports a secondary analysis of the public, de-identified data and materials of Wang et al. (2025), available at the Open Science Framework (https://osf.io/9awj3/). No new data were collected, so institutional review board approval was not required. Vocentive is the institution responsible for this work. This secondary analysis was not preregistered. Its analysis plan was recorded in a version-controlled repository before any held-out score was computed, and the plan and code are available from the author. Large language models were used under the author's direction to check analysis code and audit the manuscript against reporting standards, and the author reviewed all output and is responsible for the content. This research received no external funding. The author declares no conflicts of interest. No Head Start staff or programs were involved in the design or interpretation of this analysis.

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Abstract

Head Start staff report high psychological strain. Under conservation of resources theory, the current study tested whether intake measures predicted elevated stress and low intention to remain six months later, and whether a brief reappraisal exercise improved it. Data from a randomized trial of 4,054 employees (95% women, mean age 43) were split into training, development, and test partitions (n = 2,429, 811, and 814). An 11-predictor logistic regression on baseline stress, burnout, anxiety, depressive symptoms, job satisfaction, hope, reappraisal beliefs, and demographics predicted elevated six-month stress better than the base rate (log loss 0.532 versus 0.596, difference −0.064, 95% CI [−0.099, −0.029]). Area under the receiver operating characteristic curve was .72 [.67, .77], and the calibration slope was 0.92 [0.70, 1.19]. The same predictors predicted low intention to remain (log loss 0.386 versus 0.417, difference −0.030 [−0.060, −0.002], area under the curve .70 [.63, .77]), although the Brier difference included zero. Baseline stress and younger age had the largest standardized coefficients for stress, and lower job satisfaction had the largest for intention to remain. Adding assigned condition and post-exercise responses, or the same measures three months later, changed log loss by −0.002 [−0.006, 0.001] and −0.001 [−0.023, 0.022]. For staff who completed the follow-up, a short intake survey predicted both outcomes and the reappraisal exercise added little. After local validation, the equation could identify six months ahead which staff will likely report elevated stress or low intention to remain, while there is still time to offer support.

Keywords: emotion regulation, cognitive reappraisal, predictive validity, early childhood educators, retention

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Head Start staff experience substantial psychological strain. Among women working in Pennsylvania Head Start programs, diagnosed depression and frequent unhealthy days were more common than among U.S. women with similar sociodemographic characteristics (Whitaker et al., 2013). Across the early-childhood workforce, depressive symptoms have been higher among child care workers with low household incomes (Linnan et al., 2017) and among educators who were younger, were paid less, lacked health insurance, or held more than one job (Roberts et al., 2019). In fall 2021, Head Start teachers reported more anxiety and depressive symptoms than comparable groups surveyed before the pandemic (Harding et al., 2024). In the same year, more than half of centers reported difficulty with teacher turnover and most had trouble covering classrooms (Gonzalez et al., 2024). Wages in the field are lower than those of 98% of all other occupations (McLean et al., 2021, and Dade & McLean, 2023, as cited in Wang et al., 2025).

Brief psychological interventions have been proposed as one response to this strain (Walton & Wilson, 2018). In a recent field experiment in this workforce, 4,054 Head Start staff were randomized to a short online cognitive reappraisal exercise or to an active reflection control and were assessed again six months later (Wang et al., 2025). Relative to control, staff in the reappraisal condition reported less frequent stress ($d = -0.13$, 95% CI [−0.21, −0.05]) and stronger intentions to continue in the current organization ($d = 0.12$, 95% CI [0.04, 0.20]). The intention-to-remain measure was taken after a brief refresher exercise, so the trial authors noted that this difference may partly reflect that refresher.

A mean difference between the two conditions estimates the average effect of the exercise on each outcome. The current study used the same public data to test prediction rather

than explanation (Hofman et al., 2017; Shmueli, 2010; Yarkoni & Westfall, 2017). Specifically, the study tested how well a staff member's stress and intention to remain six months later could be predicted from baseline information, and whether a record of the exercise improved that prediction. The second outcome was measured with a single item on the likelihood of continuing to work in the current organization, which is a measure of turnover intention. Development and test participants were set aside before any model was built, and no prediction was computed for them until the analyses specified in advance were run.

Prediction as a Distinct Scientific Aim

Psychological research has focused on explaining the causes of behavior and has given far less attention to predicting it (Yarkoni & Westfall, 2017). Researchers typically specify a model from theory, estimate it in one sample, and treat the explained variance as evidence for the theory. This practice treats a model that explains well as one that will also predict well, although explanatory power does not imply predictive power (Shmueli, 2010; Yarkoni & Westfall, 2017). Explanatory models are evaluated by the interpretability of their coefficients, whereas predictive models are evaluated by their accuracy in cases that were not used to estimate them (Breiman, 2001b; Shmueli, 2010). An explanatory model that fits chance patterns in its own sample can therefore predict new cases poorly (Shmueli, 2010). In social research, a close fit in the estimation sample does not show how much of a phenomenon the theory predicts in new cases (Hofman et al., 2017). Putka et al. (2018) introduced modern prediction methods to organizational researchers and illustrated their potential to improve prediction over regression, and out-of-sample studies have appeared on turnover (Sajjadiani et al., 2019) and on personnel decisions more broadly (Campion et al., 2024; Koenig et al., 2023).

Recent evidence suggests that the expected level of prediction is modest. In a mass collaboration, 160 teams were given rich longitudinal data on the same families. About a fifth of the variance in the best-predicted adolescent outcome was explained, and simple benchmark models predicted nearly as well as the most elaborate submissions (Salganik et al., 2020). In clinical prediction studies at low risk of bias, machine-learning methods have shown no discrimination advantage over logistic regression (Christodoulou et al., 2019). Together these findings set modest expectations for prediction in the current study and suggest that flexible methods add little when a short logistic model already predicts about as well as the data allow. Two questions follow for the current study. The first is how much a prediction from baseline measures improves on the base rate and whether a short logistic model captures that improvement. The second is whether assigned condition and the immediate post-session scores add to it.

Several reporting standards follow from the decision to evaluate prediction in new cases. The cases used to evaluate a model must be separate from those used to build and tune it, because any score computed on cases that influenced model choice is optimistic (Cawley & Talbot, 2010; Varma & Simon, 2006). Evaluating the same held-out cases more than once has the same effect, because each evaluation can influence later modeling choices (Dwork et al., 2015). Because the models in the current study produce probabilities, their performance depends on calibration as well as discrimination (Van Calster et al., 2019). Log loss and the Brier score were used as the main performance measures because both evaluate the accuracy of the predicted probabilities themselves, and both are proper scoring rules (Gneiting & Raftery, 2007; Steyerberg et al., 2010). The report follows the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis statement for studies using regression or

machine-learning methods (TRIPOD+AI; Collins et al., 2024). It describes the data partitioning and every model-building step, including hyperparameter tuning, and reports calibration as a performance measure.

A Resource Account of Persistence Over Six Months

A predictive hypothesis requires a reason to expect that a staff member's state at baseline will predict that person's state six months later. Conservation of resources (COR) theory provides that reason (Hobfoll, 1989; Hobfoll et al., 2018). In the current study the theory was used to select which predictors to measure and when to measure them, and the predictive results were not treated as a test of its mechanism. People strive to retain, protect, and build resources, and stress follows the threat or the actual loss of those resources (Hobfoll, 1989). Resources are the objects, personal characteristics, conditions, and energies that a person values or uses to obtain what is valued (Hobfoll, 1989). Two principles of the theory matter here. Under the primacy of loss, resource loss is disproportionately more salient than resource gain (Hobfoll et al., 2018), so a person who is depleted does not return to baseline merely because demands pause. Under the loss spiral, those who lack resources are most vulnerable to further loss and least able to offset it (Hobfoll, 1989; Hobfoll et al., 2018). Baseline stress, burnout, anxiety, and depressive symptoms were therefore expected to predict later stress because they indicate the resources a staff member has available to meet the demands of the next six months. This expectation assumes that pay, staffing, and the emotional labor of the classroom change little over that interval. A prediction of this kind would also have practical value for a program. Intake measures are inexpensive to collect, and a validated prediction from them would indicate which staff are most likely to report elevated stress or low intention to remain six months later, while

there is still time to offer support. The prediction would not identify the cause of a staff member's risk.

Working conditions such as pay, staffing, and classroom demands can deplete or restore the resources staff have available. COR theory describes these conditions as resource caravan passageways, the environmental conditions that support or block the creation and maintenance of individual resources (Hobfoll et al., 2018). In Head Start those conditions changed slowly over the study interval. Most programs had increased wages by 2021–2022 (Gonzalez et al., 2024), yet Head Start educators in another sample still named poor benefits, compensation, and classroom-management stress among reasons for considering leaving (Zhao et al., 2025). When working conditions change little, the resources they support should also change little. Burnout has accordingly been described as a relatively continuous state, with an average rank-order stability of .56 over six months (Mäkikangas & Kinnunen, 2016). In trajectory studies, with lags of typically one or two years, most employees followed a stable path, whether at a low, moderate, or high level of burnout, and increasing, decreasing, or curvilinear change was atypical (Mäkikangas & Kinnunen, 2016). In the same review, an earlier burnout score explained less than half of the variance in a later burnout score (Mäkikangas & Kinnunen, 2016). Burnout is therefore stable in rank order and still changes within individuals. Among Head Start teachers, average anxiety and depressive symptoms decreased from fall 2021 to spring 2022 as pandemic conditions eased (Harding et al., 2024). A baseline score should therefore account for much of the six-month prediction, and a substantial share of the variance in later stress should remain unpredicted.

Under this reasoning, the set of depletion indicators at baseline is the best available summary of a staff member's resources. The appropriate test is therefore whether a model using

those indicators predicts better than the base rate under proper scoring rules (Gneiting & Raftery, 2007; Salganik et al., 2020; Steyerberg et al., 2010). A test of one coefficient, such as burnout net of anxiety, would answer a different question, because predictive success is defined by error in cases the model has not seen and not by the size of a single coefficient (Yarkoni & Westfall, 2017). A highly significant effect can leave outcomes for new observations largely unpredictable (Rocca & Yarkoni, 2021). The expected improvement over the base rate was modest. Rich data and methods designed for prediction have produced predictions of life outcomes that were only slightly better than a simple benchmark model (Salganik et al., 2020). In this workforce, some depleted staff will recover and some well-resourced staff will become depleted.

Hypothesis 1. For elevated six-month stress, the baseline-only logistic model retained by the pre-declared development rule and fit on the training partition will have lower test-partition log loss and Brier score than an intercept-only model that predicts the training base rate.

Job Attitudes, Exhaustion, and the Intention to Remain

The second outcome is the intention to continue working in the current organization. Turnover intention is the strongest proximal antecedent of voluntary turnover (Hom et al., 2017), and job satisfaction predicts it more strongly than organizational commitment does (Tett & Meyer, 1993). Under COR theory, both job satisfaction and the depletion indicators were expected to predict intention to remain. Under the desperation principle, people whose resources are outstretched or exhausted enter a defensive mode to preserve the self (Hobfoll et al., 2018). Withdrawal from the job is one such defense. Burnout is a prolonged response to chronic interpersonal stressors on the job, and exhaustion, the draining of emotional resources, is one of its three key dimensions (Maslach & Leiter, 2016; Mäkikangas & Kinnunen, 2016). Job satisfaction enters the classical turnover models as a proximal antecedent of search and quit

intentions (Hom et al., 2017). Exhaustion indicates the staff member's remaining emotional resources, and job satisfaction indicates the staff member's evaluation of the job. Both were therefore expected to predict intention to remain six months later. In a recent Head Start sample, emotional exhaustion was associated with intentions to leave the current program and position, and compassion satisfaction with intentions to leave the program and the profession (Zhao et al., 2025). The change trajectory of job satisfaction also explains variance in quit propensity beyond its static level (Hom et al., 2017), a point taken up under Hypothesis 4.

Low intention to remain was uncommon in the sample, roughly one staff member in eight, so the informative comparison is with the base rate under proper scoring rules, and the precision–recall curve is used as the discrimination summary. The outcome is a stated intention, which the turnover literature places upstream of turnover (Hom et al., 2017).

Hypothesis 2. The baseline-only logistic model retained by the same rule for this label will have lower log loss and lower Brier score on the test partition than the intercept-only model, for low six-month intention to remain.

Incremental Value of a Brief Intervention

Prior reappraisal research does not indicate whether a record of one brief session should improve the prediction of stress six months later. In the process model of emotion regulation, reappraisal is a form of cognitive change, one of the antecedent-focused strategies that act before the emotion is generated (Gross, 1998). It cognitively transforms a situation so as to alter its emotional impact (Gross, 1998). Meta-analytic evidence puts the effect of cognitive change on emotional outcomes at $d = 0.36$ (Webb et al., 2012). The extended process model distinguishes three stages at which regulation can succeed or fail (identification, selection, and implementation; Gross, 2015). A fourth stage, monitoring whether to stop regulating or to switch

strategies, has since been added (Sheppes et al., 2015). A single online session targets the selection stage. It can raise beliefs that reappraisal works and intentions to use it, and the trial's immediate manipulation check showed that it did (Wang et al., 2025). Whether staff then implement the strategy over the following six months is a separate question, because the exercise does not change the classrooms in which they work.

A brief intervention has a lasting effect only when the new meaning it introduces changes the person's behavior and the situations the person then enters reinforce that meaning (Walton & Wilson, 2018). Whether that process continues depends on the person and the setting, so any benefit from a single session should differ across staff (Walton & Wilson, 2018). In a four-session laboratory study, distancing but not reinterpretation was associated with lower perceived stress in daily life, a difference that emerged within 2 to 5 days after training began (Denny & Ochsner, 2014). An eight-week workplace program of digital cognitive-behavioral and emotion-regulation skills did not show a statistically significant difference between conditions in a feasibility trial (Moukhtarian et al., 2024). Who benefits also depends on the circumstances of the stressor. Reappraisal's benefit is clearer when adversity is relatively uncontrollable (Mauss & Troy, in press). Cognitive reappraisal ability was unrelated to depressive symptoms at low stress but associated with fewer symptoms at high stress (Troy et al., 2010).

Two further considerations concern how much the session should add. First, if the six-month effect of assignment were homogeneous and of the size the trial reported, assignment would shift each person's probability of elevated stress by only a few percentage points, and knowing the assignment would reduce log loss very little. Second, the mediator the source trial identified was the proportion of writing-task words on the reappraisal topic. The preregistered mediators, beliefs and intentions, were not supported (Wang et al., 2025). There is therefore no

basis in the mediation result for expecting post-session beliefs, intentions, hope, job satisfaction, or anticipated resilience to add predictive validity above and beyond assignment itself. Hypothesis 3 nevertheless tests the incremental contribution of assigned condition together with the five immediate post-session scores, referred to here as the session measures, because the practical question is whether any record of the exercise improves the prediction.

The resource reasoning also leads to a comparative prediction, which ranks two effects by magnitude and is more precise than a prediction of direction alone (Edwards & Berry, 2010). It is expected that the reduction in log loss from the baseline predictors, relative to the base rate, will exceed the further reduction from adding the session measures, with both improvements computed in the same elastic-net class. The baseline measures summarize resources expected to persist over six months, whereas the session measures record a single session that acted on strategy selection. An improvement from the session measures as large as the baseline improvement would weaken this reasoning.

Hypothesis 3. For elevated six-month stress, adding assigned condition and the immediate post-session scores to the baseline elastic-net specification will lower test-partition log loss relative to that baseline elastic-net specification.

Incremental Value of a Midpoint Measurement

The trial’s midpoint booster session, administered roughly midway between baseline and follow-up, opened with check-in questions, followed by a refresher section and post-refresher questions (Wang et al., 2025). The check-in questions re-measured stress, burnout, happiness, reappraisal beliefs and intentions, job satisfaction, and resilience. The booster registration specifies measurement immediately before and after the refresher (Wang, 2023a). Under COR theory, the direction of change between baseline and midpoint is informative in itself, because a

staff member on a loss spiral and a staff member who is recovering can have the same baseline score and then move in opposite directions. The parallel argument for attitudes is that change in job satisfaction predicts leaving beyond its level (Hom et al., 2017). The midpoint measures were therefore expected to improve the prediction beyond the baseline and session measures.

Hypothesis 4. Among participants with midpoint data, adding the pre-refresher midpoint measures to the Hypothesis 3 model, refit on the training participants who had midpoint data, will lower test-partition log loss further on that subsample.

Scope Conditions

The hypotheses above were stated for one setting. They concern one U.S. human-service workforce during the 2022–2023 program year, followed for roughly six months with single-item and two-item self-report measures. They predict that strain and attitudes will persist, that prediction from baseline will be moderate, and that a brief exercise will add little and will not benefit all staff equally. They do not predict that the models will transfer to other workforces or time spans without external validation (Rocca & Yarkoni, 2021; Salganik et al., 2020), and the causal effect of reappraisal is left to the source trial.

The Present Study

This study analyzed public Study 2 data from the Head Start reappraisal field experiment (Wang et al., 2025; https://osf.io/9awj3/). Models were built and tuned in a person-level training partition, compared once on a development partition, and scored once on a held-out test partition after the model was fixed. The four hypotheses were fixed before any held-out prediction or score was computed. Hypotheses 1 and 2 concern prediction from baseline alone, and Hypotheses 3 and 4 concern the added value of the session measures and of the midpoint measures. Analyses of the source exclusion-rule sample and of burnout as an alternate label are

reported as supplementary checks. A wider set of model classes is reported as internal validation, and attrition is described in the preliminary analyses (Table S6).

Method

Source Experiment

The source study was a two-condition randomized field experiment with Head Start employees (Wang et al., 2025). The registration admitted Head Start and Early Head Start staff in the United States. The achieved sample included communities in every U.S. state and in Puerto Rico and the U.S. Virgin Islands (Wang, 2022; Wang et al., 2025). Recruitment was online, through program leaders, state and national Head Start associations, the MyPeers platform, social media, and the National Head Start Association newsletter. The registration set no age, hours, or role criterion beyond employment as Head Start or Early Head Start staff, and the preregistered exclusions were data-quality rules rather than eligibility screens (Wang et al., 2025). The study link was open from 6 September to 23 December 2022. Participants were randomized to a brief online cognitive reappraisal exercise or an active reflection control, in English or Spanish. A midpoint session (17 January–12 February 2023) and a six-month session (6 April–15 May 2023) repeated core measures. The original authors preregistered the initial session and each follow-up (Wang, 2022, 2023a, 2023b). Immediately after the session, on the unexcluded sample of 4,054, the reappraisal condition reported stronger beliefs that reappraisal works than the reflection condition (d = 0.08, 95% CI [0.02, 0.14]; Wang et al., 2025). The reappraisal condition also reported stronger intentions to use it (d = 0.19, 95% CI [0.13, 0.25]). A wrong answer on the video comprehension items produced a clarification on screen, and the participant was retained. The source report describes attention items and set-the-slider checks but no bot, CAPTCHA, or duplicate-response screen, and it gives the removal under its preregistered

data-quality rules as a pooled 38.63% rather than by stage. This analysis kept all 4,054 participants and applied that rule as a sensitivity analysis (Supplementary Analyses; Table S11). The report does not give the number of participating programs or sites. Participants in the source trial completed a consent form at the start of the initial session (Wang et al., 2025). The preprint does not name the reviewing board. The present secondary analysis used only the public de-identified file and involved no contact with participants. Outcomes were self-reports at the six-month session. Participants were not blinded to the exercise they had completed, and condition entered the Hypothesis 3 specification as a predictor.

Participants

Participants were Head Start employees in the United States, Puerto Rico, and the U.S. Virgin Islands (the registration also admitted Early Head Start staff). The combined person file contains 4,054 participants with a baseline record (reappraisal n = 2,002; reflection n = 2,052). Median follow-up was 184 days (M = 180.76, SD = 28.11, range 106–248), with 42% attrition from baseline to six months as reported by the trial and no reliable difference between conditions (Wang et al., 2025). In this frame, 2,347 participants had a numeric six-month stress item and 2,183 a numeric intention-to-remain item. In the training partition (n = 2,429), age averaged 42.92 years (SD = 12.46; range 18–83). Of those reporting gender, 94.8% were women. Education used a seven-level code. The most common levels were a bachelor's degree (34.0%), an associate's degree (21.7%), and some college (15.9%). The most common positions were preschool education (31.4%) and infant and toddler education (13.6%). Management or administration and a multiple-position response each accounted for 13.3%. Most participants were affiliated with a Head Start or Early Head Start program (96.5%) and completed the survey in English (96.0%). On the race and ethnicity item, 2,424 training participants responded and

could select all that apply. Of those, 50.0% selected White, 24.3% Hispanic or Latino/a, and 19.5% Black or African American. Smaller shares selected American Indian or Alaska Native (1.7%), Multiracial (1.4%), Pacific Islander (1.0%), and Middle Eastern or North African (1.0%). Further shares selected East Asian (0.8%), Southeastern Asian (0.7%), South Asian (0.4%), and another identity (1.5%). Another 3.2% preferred not to say, and 5.2% selected more than one option, so the percentages sum to more than 100 (Table S8b). Race and ethnicity were not entered in any model, as predictors or as covariates, because the predictor set was fixed before any held-out analysis and because a prediction intended for program use should not condition on a protected characteristic (see Limitations). They are reported here to describe the sample. Baseline predictor means and six-month outcome rates by gender, survey language, and race and ethnicity in the training partition are in Table S18. The file contains no measure of tenure or years of experience. The items that record the month and year a person first and last worked in Head Start were not used. Training-partition correlations are in Table S1. Table 1 compares the three partitions on the short-model predictors, assigned condition, and the two outcomes. Continuous predictors differed by at most 0.06 pooled standard deviations. Outcome prevalence was not balanced by the stratification (baseline stress × condition). Elevated six-month stress was 31.9% in training versus 26.8% and 27.9% in the held-out partitions, which is why the calibration intercepts below are shifted from zero. Figure 1 shows the flow from the combined file to the partitions and labeled subsets.

Person-Level Partition

A held-out partition estimates performance without bias only if no score computed on it influenced the choice of model (Cawley & Talbot, 2010; Varma & Simon, 2006). Evaluating the same holdout more than once removes that property (Dwork et al., 2015). Participants were

therefore assigned once to training, development, or test partitions in a 60/20/20 ratio. The assignment was stratified on assigned condition crossed with baseline stress, so that each partition held approximately the same proportion of each stratum (Krstajic et al., 2014). The partitions contain 2,429, 811, and 814 participants. Among those with a numeric six-month stress label the counts are 1,411, 474, and 462. Among those with a numeric intention-to-remain label they are 1,315, 438, and 430, and among those with both midpoint and six-month stress, 1,143, 371, and 384. The assignment was not altered after creation, and the label cuts are scale anchors rather than sample quantiles. The block-assignment rule and the random seeds are in Supplement S3 and in the analysis code available from the author (Heil et al., 2021). The split script read the full public frame once to assign partitions and wrote an inventory of aggregates. The inventory held row counts per partition, counts of numeric six-month responses per partition, and full-sample event counts and prevalence for the stress, intention, and burnout labels. Those aggregates informed the choice of stress as the primary label, the decision to treat intention to remain as secondary, and the precision statements below. They could not inform model form, features, penalties, or candidate choice, which were fixed on training rows. The claim is therefore the narrower one, that no development or test prediction was computed before the authorized analyses were run. No development or test prediction or performance score was computed before the authorized analyses, and the confirmatory protocol was fixed after that inventory and before any such scoring.

Measures

All items are from the original instruments archived at the Open Science Framework (OSF) project. Variable names follow the original authors' combined file. Response scales for the modeled items are given below and in the note to Table S1. Internal consistency for the three

two-item composites (anxiety, depressive symptoms, job satisfaction) was estimated in the training partition from the item-level source file as the Spearman–Brown coefficient and is reported on the diagonal of Table S1. Single items have no internal-consistency estimate. Apart from the PHQ-2 and GAD-2 screeners, the items were written by the original authors (Wang et al., 2025).

Outcomes (Six-Month Session)

Elevated stress was defined as a response of 3 or 4 on the single item "Feeling stressed," rated over the previous two weeks. The response scale was a 4-point frequency scale (1 = *not at all*, 4 = *nearly every day*). The cut corresponds to *more than half the days* or *nearly every day* and was fixed from the instrument before analysis. Low intention to remain was defined as a response of 50 or lower on the item "How likely are you to continue working in your current organization in the next 12 months?" The anchors were 0 = *not at all* and 100 = *certainly*. The original authors coalesced the item across its two routing versions. Burnout used the item "Feeling burned out" on the same 4-point scale, with the cut at 3, as an alternate label in the supplementary analyses. The six-month items were administered online in English or Spanish with the same wording in both conditions. The source report does not describe how the Spanish version was produced, and no translation or back-translation record is available to this secondary analysis. In the training partition, 4.0% of participants completed the survey in Spanish. Participants received a $20 gift card for the six-month session and $15 for the midpoint session, with a $100 lottery after each activity (Wang et al., 2025).

Baseline Predictors (Initial Session, Pre-Exercise)

Stress and burnout were the single items quoted under Outcomes, on the 4-point frequency scale (1 = *not at all*, 4 = *nearly every day*). Happiness was the single item "Feeling

happy" on that scale. Depressive symptoms were the mean of the PHQ-2 items "Having little interest or pleasure in doing things" and "Feeling down, depressed, or hopeless" (Kroenke et al., 2003), both on the same 4-point frequency scale, with a training-partition Spearman–Brown coefficient of .71. Anxiety was the mean of the GAD-2 items "Feeling nervous, anxious, or on edge" and "Not being able to stop or control worrying" (Kroenke et al., 2007), both on the same 4-point frequency scale, with a training-partition Spearman–Brown coefficient of .77. Hope was the single item "Right now, I'm feeling hopeful, optimistic, and encouraged" on the 7-point agreement scale (1 = *strongly disagree*, 7 = *strongly agree*). Job satisfaction was the mean of "Right now, I'm feeling satisfied with my job" and a reverse-scored "Right now, I'm feeling negative about my job." The scale was 7-point agreement, and the training-partition Spearman–Brown coefficient was .79. The raw correlation between the two items was $r = -.65$, consistent with the reverse scoring. Workplace resilience was the single slider item "In the last two weeks, to what extent have you been able to bounce back after illness or hardship in the workplace?" (0 = *0% of the time*, 100 = *100% of the time*). Habitual reappraisal was measured by three 7-point agreement items. They were "In the last two weeks, I've tried changing how I think about a situation in order to feel better" (behavior) and "I believe that changing how I think about a situation can change my emotions" (beliefs). The third was "I plan to change how I think about a situation when I want to change my emotions" (intentions). Demographics were age in years, gender (female or not female, which the preprocessing pipeline coded as two indicator columns), education, interview language, program affiliation, and job position. Education was a seven-level code whose levels were some high school or less, high school graduate, general equivalency diploma, some college (no degree), associate's degree or equivalent, bachelor's degree or equivalent, and graduate or professional degree. The response "I prefer not to say" was set to

missing. Job position was coded as educator, management or administration, or a mutually exclusive position category. Affiliation contrasted a Head Start or Early Head Start program with another employer.

Post-Session and Midpoint Predictors (Hypotheses 3 and 4)

Predictors added for Hypothesis 3 were assigned condition and the immediate post-exercise scores. Those scores were reappraisal beliefs and intentions, hope, job satisfaction, and anticipated workplace resilience. Anticipated resilience used the same item phrased for the next two weeks ("In the next two weeks, to what extent do you think you will be able to bounce back after illness or hardship in the workplace?"). For Hypothesis 4, the midpoint booster measures were the check-in questions answered before the refresher section (Wang et al., 2025). They were stress, burnout, happiness, reappraisal beliefs and intentions, job satisfaction, and resilience, identified by the original authors' pre-refresher variable names. Midpoint measures collected after the refresher were not used.

Excluded as Predictors

Any six-month measure other than the label, free-text responses, word counts, the original authors' full-sample topic-model proportions, page timing variables, change scores that embed a post-session value, exclusion flags, and the six-month extraversion items.

Analytic Strategy

The person is the analytic unit, and every number reported for the development or test partitions comes from a model whose parameters and preprocessing (imputation values, scaling constants, one-hot coding, penalty) were determined without those participants. For Hypotheses 1 and 2, four pre-declared candidates were compared on the development partition and the retained candidate was scored on the test partition against the intercept. The first candidate was

an intercept-only model that predicts the training base rate. The second was an unpenalized logistic regression on a short, pre-declared baseline set of 11 predictors (stress, burnout, anxiety, depression, job satisfaction, reappraisal beliefs and intentions, hope, age, education, and female). Anxiety, depression, and job satisfaction are two-item means, so the 11 predictors draw on 11 questionnaire items and three demographic fields. Gender entered the fitted equation as two indicator columns (female, not female), so the 11 predictors occupy 12 coefficient columns (Table S12). The third was a penalized logistic regression on the full baseline predictor list, with an elastic-net penalty spanning the ridge and lasso endpoints (Zou & Hastie, 2005). The fourth was an equal-weight probability average of the short logistic, the elastic net, a random forest (Breiman, 2001a), and gradient boosting (Friedman, 2001). The average used equal weights declared in advance, and no second-level model was fit to combine the candidates. For Hypotheses 3 and 4, the elastic-net specification with the added predictors was compared with the elastic-net specification without them, on the same test rows. For Hypothesis 4, the comparator is the Hypothesis 3 feature set refit as an elastic net on the midpoint-complete training rows (n = 1,143), so that both specifications share identical training rows. Logistic regression was retained as the model class on the strength of systematic-review evidence that more flexible learners do not improve discrimination over logistic regression in low-bias comparisons (Christodoulou et al., 2019). A wider set of model classes, from intercept to gradient boosting and a stacked ensemble, was evaluated inside the training partition only and is reported as internal validation.

Missing predictor values were imputed within each training fold (median for numeric, mode for categorical), followed by standardization and one-hot coding fit on the same fold, so that no preprocessing step saw the held-out rows (Kapoor & Narayanan, 2023). Category levels

that did not appear in a training fold were ignored and were not given a new column. Median imputation is a simple rule whose use in applied work has been advised against (Pargent et al., 2023). It was retained here because the three predictors with any missing values (age, education, and the gender item) were each missing for 2.1% or fewer of training participants (Table 1 and Table S8). Those imputations treat the missing values as missing at random given the observed predictors in that fold. Missingness indicators were not added as predictors, because the short predictor set was fixed before any held-out analysis. Participants without a numeric label for a given outcome were not scored for that outcome, so the confirmatory metrics describe predictions for staff who completed the six-month survey. Attrition is described in the preliminary analyses and was not weighted into those metrics, because a weighting model would itself be a fitted object whose misspecification could not be checked on the held-out partitions. A training-only inverse-probability-weighting sensitivity, added after outside review, is reported in Supplement S15.

No outlier screen was applied, and no value was winsorized or trimmed. Survey responses lie on fixed scales, and the only out-of-range entries were non-numeric tokens, which were treated as missing. The two labels are imbalanced (31.9% elevated stress and 11.4% low intention to remain among labeled training rows), and no oversampling, undersampling, or class weighting was used. The imbalance is addressed by scoring with log loss, Brier score, and the precision–recall curve, and accuracy at a threshold is not used. Imputation, coding, and the label definitions were applied identically to every participant regardless of survey language or demographic group. No fairness-specific procedure was applied, and performance was not estimated within sociodemographic groups (see Limitations). Clustering of staff within centers was not modeled. The predictions are person-level. No development sample-size calculation of

the kind proposed by Riley et al. (2020) was performed. The training partition was the trial's available sample under the 60/20/20 split, with 450 and 150 training events for the two labels. That is about 41 and 14 events per predictor for the 11-predictor short logistic model and about 24 and 8 for the 19-predictor elastic net. Supplement S12 reports the retained short logistic model for both labels (Table S12) and the full-baseline elastic net for both labels (Table S13). The training imputation values, means, and standard deviations used to score a new case are printed with those tables. A new staff member is scored by preparing those named baseline items, reusing the printed training imputation values for any missing predictor, standardizing with the printed means and *SD*s, and applying the printed coefficients. Out-of-range values are not winsorized. The stress column of Table S13 is the Hypothesis 3 comparator. Coefficients for the session- and midpoint-augmented specifications are not tabulated and are available from the author.

Penalty Selection and Internal Validation

The elastic-net penalty (inverse regularization strength C on a 10-point logarithmic grid from 0.0001 to 100; mixing parameter in {0, 0.5, 1}) was chosen by stratified 5-fold cross-validation on the training partition, minimizing mean log loss. A first path that stopped at C = 0.01 placed the stress penalty on that boundary, so the grid was extended once toward the null model before the reported fit. The selected penalty was refit on all training rows. That object is the elastic-net candidate in the development comparison for Hypotheses 1 and 2 and the specification scored for Hypotheses 3 and 4. Internal performance was estimated by nested cross-validation (five outer folds, the full penalty search repeated inside each), which avoids the optimism of reporting the same cross-validated score that selected the penalty (Cawley & Talbot, 2010; Varma & Simon, 2006). Internal validation of this kind estimates the performance of the

fitting procedure, not of the single fitted model (Bates et al., 2024). The single confirmatory score on the test partition estimates the latter. The extended model comparison used repeated stratified 5 × 3 cross-validation with nested tuning (Krstajic et al., 2014) and paired fold-wise comparisons against the short logistic model.

Development and Test

The development partition was analyzed once, for the candidate comparison. After the confirmatory analysis, a second and separately pre-specified analysis of the development and test partitions produced the cross-partition descriptives (Table 1, Figure 1) and the supplementary analyses. The probability average was chosen over the stacked model because it has no second-level parameters to tune and because the stack's training calibration slope on the intention label was 1.37. Training calibration slopes for both ensembles are in Table S4. For each label, the short logistic model was retained unless a challenger lowered development log loss by more than one paired bootstrap standard error of the difference (2,000 resamples of development participants). If two challengers cleared that margin, the lower log loss was taken, and if those two were within their own paired standard error, the elastic net was preferred to the ensemble. The intercept would have replaced the short logistic reference had it lowered log loss by more than one paired standard error. The margin is a parsimony rule analogous to the one-standard-error rule for choosing along a complexity path (Hastie et al., 2009, Section 7.10). It is set at one rather than more standard errors because larger parsimony penalties tend to worsen out-of-sample prediction relative to the minimum (Chen & Yang, 2021). All candidates' development scores are reported. The chosen model, as fit on the training partition, was then scored on the test partition in a single confirmatory analysis. It was not refit on training plus development, because refitting changes the estimand of the split (Bates et al., 2024).

Performance Measures

Log loss and Brier score are the headline comparisons for all hypotheses, because both are proper scoring rules that reward accurate probabilities (Gneiting & Raftery, 2007; Steyerberg et al., 2010). Predicted probabilities were clipped to the interval [.000001, .999999] before any score was computed. The clip matters for the tree-based components, which can emit 0 or 1, and not for the logistic models. The scores are reported with the calibration intercept and calibration slope, estimated by logistic recalibration of the outcome on the model's logit (Van Calster et al., 2019). After outside review, a third and separately pre-specified analysis of the test partition added grouped calibration (10 quantile deciles and 10 equal-width bins per model, with Wilson intervals; Brown et al., 2001; Figure S4, Tables S16a and S16b). Only bin-level aggregates were returned, and no model was refit. Discrimination is summarized by the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC). AUPRC is emphasized for the low-prevalence intention-to-remain label, because the precision–recall plot is more informative than the receiver operating characteristic plot when the classes are imbalanced (Saito & Rehmsmeier, 2015). No misclassification costs were declared, so no threshold was tuned and accuracy was not treated as a result. Bootstrap 95% percentile intervals for test-set metrics and for every paired difference were computed by resampling test participants (B = 2,000; Raschka, 2018), with both models in a comparison scored on the same resampled participants. A hypothesis was reported as supported when its stated point comparison held (lower log loss than the comparator; for Hypotheses 1 and 2, also lower Brier score) and the paired interval for the log-loss difference lay entirely below zero. The comparative expectation from the Hypothesis 3 section was evaluated descriptively from one joint resample of the test participants, with both improvements computed on each resample. That expectation was that the

improvement of the baseline elastic net over the base rate exceeds the further improvement from assigned condition and the immediate post-session scores. This criterion was recorded before any development score was computed. No correction for multiple hypotheses was applied. The four hypotheses address different models or predictor sets, each was judged on its own pre-declared interval, and the two that share a label (Hypotheses 1 and 3) are reported with their paired comparison. The number and spacing of measurement occasions were those of the source trial. The six-month horizon was adopted because it is the trial's final follow-up and the interval over which baseline standing was expected to predict later stress under the resource reasoning in the Introduction.

Anticipated Precision, Stability, and Sensitivity

The test partition was fixed by the split, so its size was not chosen for precision. Instead, the precision it affords was calculated before the held-out analyses using the training event proportions and the frozen training fit (Riley et al., 2021; Supplement S10). With 462 test participants for the stress label, the standard error of the log of the observed-to-expected ratio was expected to be about 0.07 and that of the calibration slope about 0.14. The corresponding figures for intention to remain, at about 441 participants, were 0.13 and 0.20. A calibration-slope interval of the recommended width (Riley et al., 2021) would require roughly 3,500 and 6,500 participants, so the test slopes are reported with their intervals and treated as imprecise. The stability of the baseline elastic net under resampling of the training partition was examined with the bootstrap procedure of Riley and Collins (2023; Supplement S5). Two further model classes were added to the training comparison after the candidate list had been fixed, as a sensitivity analysis. They were an approximate Bayesian logistic regression with weakly informative

Cauchy priors (Gelman et al., 2008) and an explainable boosting machine (Lou et al., 2013; Nori et al., 2019). They did not alter the candidates (Supplement S9).

Training-Only Follow-Ups After the Confirmatory Analysis

After the confirmatory scores were written, four analyses were run on the retained short logistic model using only training rows (Supplement S14; Pargent et al., 2023). The first two were apparent (resubstitution) performance, as a contrast with nested cross-validation (Austin & Steyerberg, 2017), and learning curves with the training side of each fold reduced to 25%, 50%, 75%, and 100% of its size (Raschka, 2018). The other two were leave-one-predictor refits scored on the same folds and a ten-bin out-of-fold reliability table (Van Calster et al., 2019). Fold-to-fold variation for the training comparison is in Table S4. Approximate bootstrap intervals (Riley & Collins, 2023) are in Table S4b for the linear models only, because resampling with replacement places duplicated participants on both sides of the inner folds and inflated the out-of-fold metrics of the tree-based and ensemble models. Decision-curve analysis was not run because no threshold was declared (Steyerberg et al., 2010), and permutation or Shapley importances were not needed for an unpenalized logistic equation whose coefficients are in Table S12. None of the follow-up analyses used the development or test partitions.

Transparency and Openness

The sampling plan, all data exclusions, all manipulations, and all measures in the study are described, and the report follows the APA Journal Article Reporting Standards for quantitative research and the TRIPOD+AI statement (Collins et al., 2024). The data, instruments, preregistrations, and the original authors' code are public at the Open Science Framework (https://osf.io/9awj3/; Wang et al., 2025). Respondent-level files from that archive are not redistributed here. The source trial was preregistered (Wang, 2022, 2023a, 2023b). This

secondary analysis was not preregistered in a public registry. Its analysis plan (split, candidates, penalty grid, decision rule, and hypotheses) was recorded in a version-controlled repository before any held-out score was computed. The development partition was analyzed once for the candidate comparison and the test partition once for the confirmatory analysis, each after the plan and model bundle were fixed and hash-verified. One further pre-specified analysis, run after the confirmatory analysis, produced the post-test descriptives and supplementary analyses. A third analysis of the test partition only, for grouped calibration curves and additional interval constructions requested at outside review, was specified after the confirmatory analysis, reviewed under the same procedure as the earlier analyses, and run once (Supplement S16 and S17). A fourth analysis of the test partition only returned gridded receiver operating characteristic and precision–recall curve coordinates for the final model with pointwise bootstrap bands. It was specified after the manuscript was otherwise complete, reviewed under the same gate procedure, and run once. It refit nothing and returned no per-participant values. In total the development partition was analyzed twice and the test partition four times. Only the first test analysis was confirmatory, and the three later analyses were specified after it, returned aggregates only, and refit nothing. Data were analyzed using Python, version 3.14.2 (Python Software Foundation, 2025), scikit-learn, version 1.8.0 (Pedregosa et al., 2011), pandas, version 3.0.0 (McKinney, 2010), and statsmodels, version 0.14.6 (Seabold & Perktold, 2010). The analysis code and the locked plan are available from the author. All aggregate outputs are reported in the article and the supplemental material.

Results

Preliminary Analyses

Table S1 reports training-partition means, standard deviations, and correlations for the 11 short-model predictors and the two six-month outcomes. Baseline stress, burnout, anxiety, and depressive symptoms were strongly intercorrelated (r = .51 to .67) and each related to six-month stress (r = .30 to .42). Job satisfaction and hope related negatively to six-month stress (r = −.26 and −.33) and positively to intention to remain (r = .31 and .19). The two reappraisal items were nearly unrelated to the outcomes ($|r| \leq .13$). Six-month stress and intention to remain correlated, r = −.23.

Within the training partition, 1,411 participants (58.1%) had a numeric six-month stress item and 1,315 (54.1%) a numeric intention-to-remain item. By condition, 685 of 1,200 reappraisal and 726 of 1,229 reflection participants had a six-month stress item. Participants with a six-month outcome differed little at baseline from those without. Standardized mean differences were below 0.20 in absolute value for every baseline measure except education (0.33 for the stress contrast, 0.35 for intention to remain) and reappraisal beliefs (0.22 for both), with participants who returned having somewhat more education and stronger reappraisal beliefs. The full comparison is in Table S6.

Internal Validation on the Training Partition

Repeated nested cross-validation on the training partition compared nine model classes while repeating preprocessing and tuning within each outer training fold. Across both outcomes, every class other than the intercept tied the short logistic reference under the pre-declared descriptive tie rule. The intercept had higher log loss on both outcomes (stress mean paired difference 0.070, *SD* of the 15 differences 0.024; intention 0.026, *SD* 0.018). Figure S1 shows all

outer-fold log-loss values, and Table S4 reports the full metric summaries. Figure S2 shows the out-of-fold receiver operating characteristic and precision–recall curves of the short logistic model for both labels. Model specifications and tuning details are in Table S3.

Development Comparison

The development partition was scored once for model selection. Of the 811 development participants, 474 had a numeric six-month stress item (127 events, 26.8%) and 438 had a numeric intention-to-remain item (50 events, 11.4%). Table 2 reports each candidate against the short logistic model. For both labels, neither the elastic net nor the probability average lowered log loss by more than one paired bootstrap standard error, so the 11-predictor unpenalized logistic model was retained. The intercept did not displace it. Calibration intercepts on stress were negative, consistent with a development event rate (26.8%) below the training rate (31.9%). AUROC rounded to .74 for stress and .71 for intention to remain for all three fitted models, so their rankings of participants nearly coincided.

Confirmatory Tests

The confirmatory test analysis was run once, after the development result and the frozen model bundle had been verified. Of the 814 test participants, 462 had a numeric six-month stress item (129 events, 27.9%), 430 had a numeric intention-to-remain item (62 events, 14.4%), and 384 had both midpoint and six-month stress (105 events, 27.3%). Table 3 reports absolute test-partition performance. Figure 2 reports paired comparisons, Figure 3 calibration intercepts and slopes, and Figure S4 the grouped calibration curves. Figure 4 shows the receiver operating characteristic and precision–recall curves of the final model on the test partition, with pointwise bootstrap bands. The areas under them are the AUROC and AUPRC values in Table 3. No recalibration fit failed.

Hypothesis 1

The baseline logistic model predicted elevated six-month stress better than the training base rate. Its test log loss was 0.532 (95% CI [0.494, 0.574]) against 0.596 (95% CI [0.566, 0.627]) for the intercept, a paired difference of −0.064 (paired *SE* = 0.018, 95% CI [−0.099, −0.029]). Its Brier score was .177 [.161, .195] against .203 [.189, .218], a paired difference of −.026 [−.041, −.010]. Both point comparisons held and the log-loss interval lay below zero, so Hypothesis 1 was supported. AUROC was .72 [.67, .77], and AUPRC was .52 [.43, .61] against a test event rate of .28. The calibration slope was 0.92 [0.70, 1.19], and the calibration intercept was −0.27 [−0.49, −0.05]. Predicted probabilities were too high on average because the test event rate (27.9%) fell below the training rate (31.9%) the model was fit to, the same direction of miscalibration seen in the development partition. Observed proportions by quantile decile fell below the mean prediction in eight of 10 deciles, by .07 to .12 in the upper four (.28, .37, .43, and .63 observed against .35, .44, .55, and .72 predicted), the same direction the intercept summarizes. Every mean prediction lay inside the Wilson interval of its decile's observed proportion (Table S16b). With 46 or 47 participants per decile, these grouped estimates are descriptive. Paired differences in this section are focal model minus comparator, so negative values favor the focal model. Figure 2 plots the same quantities with the sign reversed.

Hypothesis 2

The baseline logistic model also predicted low six-month intention to remain better than the base rate, by a smaller amount. Test log loss was 0.386, 95% CI [0.323, 0.455], against 0.417 [0.350, 0.488] for the intercept, a paired difference of −0.030 [−0.060, −0.002] (*SE* = 0.015). The Brier score was .118 [.095, .142] against .124 [.099, .151], a paired difference of −.007 (*SE* = 0.004) whose interval, [−.015, .002], included zero. Hypothesis 2 was supported under the

declared criterion, which required both point comparisons to favor the model and the log-loss interval to exclude zero. The Brier interval did not exclude zero and the log-loss interval's upper bound sits close to zero, so the evidence for this label is weaker than for stress. The additional interval constructions pre-specified after outside review (three further percentile seeds, BCa, and bootstrap-t; Table S17) gave the same two results. The log-loss interval excluded zero under each construction (upper bounds −0.004 to −0.002) and the Brier interval included it (.001 to .002). These are post-confirmatory checks on interval construction, computed on the same 430 participants. The conclusion rests on the interval declared in the plan. AUROC was .70 [.63, .77] and AUPRC .26 [.19, .37] against a test event rate of .14. The calibration slope was 0.79 [0.48, 1.12], below 1, with a wide interval. The pre-run precision calculation had put the calibration-slope *SE* near 0.20 at an expected test *n* of about 441. The calibration intercept was 0.17 [−0.14, 0.45], reflecting a test event rate (14.4%) above the training rate (11.4%). Observed proportions by quantile decile ran above the mean prediction in eight of 10 deciles, by .01 to .08, the direction the positive intercept summarizes. Every mean prediction lay inside the Wilson interval of its decile's observed proportion (Table S16b). With 43 participants per decile, smaller local departures would not register.

Hypothesis 3

Adding assigned condition and the immediate post-session scores (five scores, entered together) to the baseline elastic-net specification did not reliably lower test log loss. The session-augmented elastic net scored 0.532, 95% CI [0.493, 0.571], against 0.534 [0.496, 0.574] for the baseline elastic net, a paired difference of −0.002 [−0.006, 0.001] (*SE* = 0.002). The Brier difference was −.001 [−.002, .001]. The session-augmented specification had slightly lower log loss, but the percentile interval included zero, so Hypothesis 3 was not supported. On this

partition the baseline and session-augmented specifications had AUROC .72 [.67, .77] and .72 [.67, .77], calibration slopes 1.01 [0.74, 1.30] and 1.01 [0.76, 1.31], and calibration intercepts −0.26 [−0.48, −0.05] and −0.26 [−0.49, −0.06].

The observed improvements matched the order stated with Hypothesis 3. On one joint resample of the test participants, the improvement in log loss of the baseline elastic net over the intercept was 0.062 [0.030, 0.094]. The further improvement from adding the session measures was 0.002 [−0.002, 0.006]. The difference between the two improvements was 0.060 [0.027, 0.092].

Hypothesis 4

Among the 384 test participants with midpoint data, adding the pre-refresher midpoint measures did not lower test log loss. The midpoint-augmented specification scored 0.536, 95% CI [0.485, 0.590], against 0.536 [0.490, 0.583] for the session-augmented specification refit on the midpoint-complete training rows, a paired difference of −0.001 [−0.023, 0.022] (*SE* = 0.012). The Brier difference was −.001 [−.010, .008]. Hypothesis 4 was not supported. Discrimination was similar (AUROC .72 vs. .71). The calibration slope of the midpoint-augmented specification was 0.75 [0.53, 0.99], below that of its comparator (0.89 [0.63, 1.18]), which indicates that the added midpoint predictors produced probabilities more extreme than the test outcomes warranted. Calibration intercepts were −0.29 [−0.57, −0.03] for the midpoint-augmented specification and −0.29 [−0.54, −0.04] for its comparator.

The final equations are in Table S12 and can be applied to a new staff member from the 11 baseline predictors. In the stress equation, the largest absolute standardized coefficients were 0.29 for baseline stress (*OR* = 1.33) and −0.25 for age (*OR* = 0.78). In the intention equation, the largest was −0.63 for job satisfaction (*OR* = 0.53), followed by −0.28 for age (*OR* = 0.76). The

two gender indicators are on the raw scale and are not part of this ranking. These coefficients describe the fitted equation and are not tests of unique effects.

Supplementary Analyses

Two analyses were specified after the confirmatory analysis and run in a second, separately pre-specified analysis of the development and test partitions (Supplement S7; 2,000 resamples; seeds are listed in Supplement S3). Four training-only follow-ups were run on the final model (Supplement S14). None changed a confirmatory result. The source authors' preregistered exclusion rule removed 38.0% to 40.4% of each partition (Table S11). The Hypothesis 1 result held whether the model was refit on included training rows (test log loss 0.492 vs. 0.571 for the base rate; paired difference −0.079 [−0.118, −0.038]; n = 303) or applied frozen (−0.076 [−0.117, −0.034]). Calibration intercepts were more negative than in the full test sample because the included event rate was lower (24.8% vs. 27.9%). With elevated six-month burnout as the label (24.2% of 462 test participants), the same 11 predictors beat the base rate (paired difference −0.055 [−0.091, −0.015]; AUROC .73 [.67, .78]) with calibration slopes near 0.8. In the training partition, apparent log loss was within 0.011 of the nested-cross-validation mean for both labels. Learning curves flattened once the training side of each fold reached about half its labeled size. No single predictor's removal moved log loss by more than 0.012 (job satisfaction, for intention) or 0.004 (age, for stress).

Discussion

The current study tested how well a Head Start staff member's stress and intention to remain six months later could be predicted from baseline information, and whether a record of a brief reappraisal exercise improved that prediction. Baseline self-report measures predicted both outcomes better than the base rate in a test partition that no model had seen (Hypotheses 1 and

2). Neither assigned condition with the immediate post-session responses nor the midpoint measures had a test log-loss interval that excluded zero. The paired difference was −0.002, 95% CI [−0.006, 0.001], for Hypothesis 3 and −0.001 [−0.023, 0.022] for Hypothesis 4. The improvement from the baseline measures exceeded the further improvement from the session measures, which is the order stated with Hypothesis 3. For a program, the practical result is that an 11-predictor model built from intake measures predicted stress. The elastic-net models that also used the exercise record or a second survey three months later did not lower test log loss by an amount the intervals separated from zero. Neither comparison was designed as an equivalence test, so these results do not show that the added measures hold no information. All of these results describe staff who completed the six-month survey.

Interpretation of the Four Tests

The Hypothesis 1 result is consistent with COR theory, and it is also consistent with any account in which strain is stable over six months. The test does not separate the theory from a simpler stability model. A short logistic model on 11 baseline predictors reduced test log loss by 0.064 relative to the base rate and discriminated at AUROC .72, with a calibration slope near 1. Under the resource reasoning, the depletion indicators at baseline summarize the resources available to meet the next six months of demands, on the assumption that pay, staffing, and classroom demands change little over that interval. The results matched the specific prediction that the baseline measures would improve log loss by more than the session measures, and they did not match the prediction that the midpoint measures would improve it further. The size of the improvement is consistent with burnout-trajectory studies in which stable paths were typical and long-term change was uncommon over lags of one or two years (Mäkikangas & Kinnunen, 2016). The same pattern was reported in a mass collaboration on life outcomes, in which simple

models predicted nearly as well as more elaborate ones (Salganik et al., 2020). In the pre-declared training comparison, every model class other than the intercept tied the short logistic model under the tie rule. On the development partition neither the elastic net nor the probability average cleared the one-standard-error margin (Table 2). This is the same pattern reported in low-bias comparisons of machine learning with logistic regression (Christodoulou et al., 2019).

Two supplementary checks agreed with the confirmatory stress result. The source authors' exclusion-rule sample removes about two fifths of participants for attention and writing-task failures. In that sample the baseline model's improvement over the base rate was slightly larger (−0.079), so retaining those participants in the confirmatory analyses did not make the prediction look better than it was. With burnout in place of stress as the label, the same 11 predictors reached similar discrimination (AUROC .73), which is consistent with treating stress and burnout as two indicators of the same strain.

Low intention to remain was predicted better than the base rate on log loss. The Brier point estimate favored the model, but the Brier interval included zero. The log-loss interval's upper bound was −0.002, and the calibration slope (0.79 [0.48, 1.12]) was imprecise. The conclusion follows the rule declared before the test analysis, which required the log-loss interval to exclude zero. The precision calculation run before any test analysis had anticipated this imprecision. At the anticipated test size of about 441 participants with a numeric intention label (about 50 events at the training event proportion), a slope standard error near 0.20 was expected. An interval of the recommended width (Riley et al., 2021) would have required about 6,500 test participants for intention to remain, roughly 15 times the anticipated 441 (Table S10). The outcome is a stated intention, which the turnover literature places upstream of leaving (Hom et al., 2017). In the training partition, job satisfaction had the strongest correlation with intention to

remain (r = .31), ahead of baseline burnout (−.20) and hope (.19; Table S1), so the attitude contributed more to this prediction than the depletion indicator did. COR theory treats both low satisfaction and exhaustion as reasons for withdrawal, and in this sample the staff member's evaluation of the job was the stronger predictor.

Hypothesis 3 was not supported, and the improvement from the baseline measures remained larger than the further improvement from the session measures. Adding assigned condition and the five post-session scores changed test log loss by −0.002 (SE = 0.002, 95% CI [−0.006, 0.001]). The interval excludes an improvement as large as the Hypothesis 1 difference (−0.064), which would have been detectable at this test size, and any remaining improvement is about 0.006 log loss or smaller. The source trial reported an average effect of d = −0.13, 95% CI [−0.21, −0.05], which concerns the mean difference between arms rather than individual prediction. A small increment is consistent with the process-model expectation that a single session acts at strategy selection and does not control implementation in the months that follow (Gross, 2015). Wise-intervention research leads to the same expectation, because a lasting effect depends on a recursive process that continues for some people and not others (Walton & Wilson, 2018). The order of the two improvements was stated before the test analysis, and the test sample was large enough to detect the difference between them.

Hypothesis 4 was not supported. Measures repeated roughly three months before the outcome did not improve the prediction beyond the baseline and session measures in the subsample that had them. This result does not support the expectation that a more recent measure would dominate an older one or that change would predict beyond level (Hom et al., 2017). The midpoint subsample was smaller (384 test participants) and the paired difference was −0.001 [−0.023, 0.022] (SE = 0.012), so an improvement of the size seen for Hypothesis 1 would have

been detectable but a smaller one would not. The midpoint-augmented specification's calibration slope (0.75) was lower than its comparator's (0.89), which suggests that the added midpoint predictors, tuned on 1,143 training rows, produced probabilities more extreme than the test outcomes warranted. Overfitting in the smaller training subsample is one explanation. Low precision, the reliability of single-item midpoint measures, and selection into midpoint completion are other explanations, and the design does not separate them. For a program considering a second survey, these seven midpoint items did not improve the six-month stress prediction among staff who completed that survey, so a second survey would need a justification other than predictive accuracy.

Relation to the Source Trial

Assigned condition was entered as a predictor to test whether it improved the prediction of a staff member's later stress, which is a different question from whether the exercise changed that stress. The mean difference reported by Wang et al. (2025; $d = -0.13$, 95% CI [−0.21, −0.05], $p = .002$) is a different quantity from an improvement in individual prediction, and the two results are compatible. A lower predicted probability for a participant in one condition would not have been evidence that the exercise reduced that participant's stress.

Implications for Prediction in Organizational Research

Log loss and Brier score are proper scoring rules, and the calibration intercept and slope summarize weak calibration (Gneiting & Raftery, 2007; Van Calster et al., 2019). Figure S4 gives the grouped curves, whose intervals are wide enough that they would show only a large local departure. Within that resolution, the upper stress deciles show observed proportions below the mean prediction. For example, Campion et al. (2024) reported correlations and incremental R^2, and Koenig et al. (2023) reported the area under the receiver operating characteristic curve.

Table 3 reports log loss, Brier score, calibration, and AUROC with intervals so that a reader can judge how far a predicted probability can be trusted.

The more flexible models did not outperform the unpenalized logistic model under the pre-declared rule. Every class other than the intercept tied it in the training comparison, and on the development partition neither the elastic net nor the probability average cleared the one-standard-error margin (Table 2). The 11-predictor logistic model is therefore the most parsimonious model at comparable performance, and a program can inspect and score it with ordinary spreadsheet software.

Scoring the equation in a program takes three local steps, which any prediction model requires before use. They are recalibrating the intercept, checking local calibration, and setting a decision rule, and they are set out here with the equation. Table S12 gives the equation and Table S12b the constants needed to score a new staff member from the 11 baseline predictors. The items are entered on their original response scales, standardized with the training constants, and combined in a logistic equation that a spreadsheet can compute. A missing item can be replaced by the training imputation value in Table S12b, which is what the model did during fitting. An out-of-range response should be treated as missing and replaced in the same way. Gains from more training data were small over the observed range. Out-of-fold log loss changed little once the training side of each fold reached about half its labeled size. In the leave-one-predictor analysis, removing age reduced the stress prediction most and removing job satisfaction reduced the intention prediction most, which agrees with the coefficients in Table S12. The intercept needs recalibration against local outcome data, because the base rate sets it. For elevated stress, the probabilities here ran too high on average because the test event rate fell below the training rate. Updating the intercept to a local prevalence does not replace checking the slope and local

calibration in the new setting. The program needs a decision rule, which requires the cost of a missed case relative to a false alarm and the capacity it can offer. With those stated, the prediction can be evaluated at the implied threshold with decision-analytic measures. No threshold was set here because no costs were declared. The prediction applies to staff whose six-month state would be observed.

Limitations

The outcomes are single self-report items dichotomized at instrument anchors. The cuts were fixed before analysis and correspond to the response options. Even so, a single item measured over two weeks is a coarse index of six-month strain, and dichotomizing that response discards the ordering of the response options (Sauerbrei et al., 2020). The cut also defines the target as the chosen anchor on this item and not as occupational stress in general. Discrimination was limited in part because the outcome is one dichotomized item, and the prediction cannot be more precise than that item. The intention outcome is a stated intention, and administrative turnover records would change both the base rate and the predictability of the outcome.

Of 4,054 baseline participants, 2,347 (57.9%) had a numeric six-month stress item and 2,183 (53.8%) had a numeric intention-to-remain item, so nonresponse was 42% for stress and 46% for intention. Participants who returned had somewhat more education and stronger reappraisal beliefs than those who did not (Table S6). The predictions therefore describe staff who complete a six-month survey. In a training-only sensitivity, reweighting labeled rows toward the full baseline cohort on baseline information did not change out-of-fold log loss, Brier score, or AUROC at the reported precision, and no coefficient shifted by more than 0.07 per *SD* (Supplement S15). That result holds under one fitted response model whose own misspecification is unchecked, and it does not rule out attrition bias in the test-partition

estimates. Baseline information predicted response only weakly (AUROC about .65), and if staff who left the program or stopped responding differed in ways those measures do not capture, the test scores would not transfer to them. The direction of that bias cannot be determined from these data.

The study concerns one workforce in the United States in one program year. Its participants were staff who enrolled in a well-being study, which limits the prediction to the employees a program would reach with a similar invitation. Base rates, predictor distributions, and the relationships among them may differ in other years, other countries, and other human-service settings. The test partition is a random draw from the same population as the training partition, so the results concern internal validity and do not show transfer across sites (Moons et al., 2015; Rocca & Yarkoni, 2021). A program elsewhere should expect to recalibrate the intercept to its own base rate before use.

All predictors and outcomes are self-report from the same participant. Common-method variance may inflate the correlations in Table S1 and, through them, the predictions. Whether it does depends on whether the method component is stable over six months, which these data cannot show.

Performance was not estimated within sociodemographic groups, and no group-specific AUROC, calibration slope, or log loss is reported. Of training participants who reported gender, 94.8% were women. Half identified as White, a quarter as Hispanic or Latino/a, and a fifth as Black or African American (Table S8b). That composition reflects the workforce itself, because the question concerns Head Start staff, and it still limits generalization. Generalization to workforces with a different gender, racial, or ethnic composition, or to early-childhood settings outside the United States, is not supported by these data. The 62 and 129 test events do not

support subgroup estimates. No fairness-specific procedure was applied, so whether the prediction behaves equally across groups is untested. Depressive symptoms in this workforce have varied with age and pay (Roberts et al., 2019) and with household income (Linnan et al., 2017), and age is a predictor in the equation. A program should check calibration within its own groups before acting on the prediction. Staff were nested within centers, and that clustering was not modeled. The participant-level bootstrap treats participants as independent, so its intervals may be narrow if within-center dependence is material. A model that pooled information within sites might also behave differently.

Calibration on the held-out partitions was assessed as intercept and slope (Figure 3) and, after outside review, as grouped observed-versus-predicted curves computed in a third pre-specified analysis (Figure S4, Tables S16a and S16b). Per-participant held-out predictions were not retained under the project's governance rules, so smooth calibration curves and the integrated calibration index were not computed. The grouped curves rest on 38 to 47 participants per decile, with Wilson intervals wide enough that only large local departures would register. Within that resolution the final model had no decile outside its interval on either label. Predicted stress probabilities were too high in the upper deciles, which the negative calibration intercept also indicates. One decile each for the session-augmented model and its midpoint-subsample fit fell outside (mean prediction about .43, observed .24 to .26; Table S16b). The equal-width bins above .8 (above .5 for intention to remain) held fewer than five participants, or were suppressed to protect such a bin, for every model, so calibration at high predicted risk is unexamined. Updating the intercept on the development partition before the test analysis was not part of the pre-declared plan and was not done. The plan fixed the training fit as the model under test, so the

test intercepts are reported as estimated. The test-partition intervals for the calibration slopes are also wide (Riley et al., 2021).

Analytic choices also bound the conclusions. The Hypothesis 3 and 4 comparisons were made in the elastic-net class so that the two improvements would be comparable. The retained Hypothesis 1 model was the unpenalized logistic model, and the test does not show whether adding the session measures to that model would have behaved differently. The training-only sensitivity analyses found one model class, an explainable boosting machine, that scored better than the short logistic model on the stress label (lower log loss by 0.009; Supplement S9). The Bayesian logistic model scored worse than the short logistic model on the intention label. Both sensitivity models were added after the candidate list was fixed, and neither was scored on the development or test partition.

Future Directions

An external validation of the baseline model, as fit here, in another early-childhood or human-service workforce would show whether the model transfers across sites and years and would allow the calibration slope to be estimated with the recommended precision (Riley et al., 2021). Linking survey responses to administrative turnover records would replace the intention outcome with the behavior it is meant to anticipate and permit a prediction at a horizon the program sets. A study that states the costs of a missed case and a false alarm would make it possible to evaluate the prediction at the threshold those costs imply. In that study a program would state what a missed case and a false alarm cost it and what support it could offer. Repeating the Hypothesis 4 comparison with a larger midpoint sample and with the midpoint items selected in advance would separate a null from the overfitting that the midpoint-augmented calibration slope suggests.

Conclusion

Head Start staff who will report elevated stress or low intention to remain six months later can be identified, with moderate accuracy, from the self-report items they complete at intake. Baseline strain and age contributed most to the stress prediction, and job satisfaction contributed most to the prediction of intention to remain. A record of a brief reappraisal exercise, and a second survey three months later, added little that the intake survey had not already captured. The practical result is a simple 11-predictor equation that a program can score in a spreadsheet. Its use requires validation in a new sample and an intercept set to the program's own base rate. With those steps taken, a program would know which staff are likely to be struggling months before the six-month survey would record it, while there is still time to offer support.

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Table 1

Participant and Outcome Characteristics Across the Training, Development, and Test Partitions

Characteristic	Training	Development	SMD (training − development)	Test	SMD (training − test)
Partition *n*	2,429	811	—	814	—
Baseline stress	2.36 (0.95)	2.36 (0.95)	0.00	2.36 (0.95)	0.00
Baseline burnout	2.13 (0.99)	2.14 (1.04)	−0.01	2.11 (1.00)	0.02
Anxiety (GAD-2)	2.02 (0.84)	1.98 (0.83)	0.05	2.02 (0.85)	−0.01
Depressive symptoms (PHQ-2)	1.63 (0.74)	1.62 (0.70)	0.01	1.63 (0.74)	−0.01
Job satisfaction	5.07 (1.46)	5.04 (1.53)	0.02	5.00 (1.52)	0.05
Reappraisal beliefs	5.82 (1.16)	5.82 (1.13)	0.00	5.85 (1.08)	−0.02
Reappraisal intentions	5.48 (1.13)	5.49 (1.13)	−0.01	5.46 (1.14)	0.01
Hope	5.29 (1.40)	5.27 (1.43)	0.02	5.26 (1.44)	0.02
Age (years)	42.92 (12.46)	43.46 (12.56)	−0.04	42.89 (12.41)	0.00
Female	2,296 (94.8%)	761 (94.0%)	—	777 (96.2%)	—
Education	5.03 (1.51)	5.00 (1.54)	0.02	5.11 (1.43)	−0.06
Assigned to reappraisal	1,200 (49.4%)	401 (49.4%)	—	401 (49.3%)	—
Elevated six-month stress	450 / 1,411 (31.9%)	127 / 474 (26.8%)	0.11	129 / 462 (27.9%)	0.09
Low six-month intention to remain	150 / 1,315 (11.4%)	50 / 438 (11.4%)	0.00	62 / 430 (14.4%)	−0.09

Note. Continuous characteristics are *M* (*SD*). Female and Assigned to reappraisal are *n* (%). Outcomes are events / labeled *n* (%). All baseline predictors were complete except age (valid *n* = 2,402, 804, and 803 in the training, development, and test partitions), education (2,378, 794, and 791), and the gender item (2,423, 810, and 808). GAD-2 = two-item Generalized Anxiety Disorder screener (Kroenke et al., 2007). PHQ-2 = two-item Patient Health Questionnaire depression screener (Kroenke et al., 2003). SMD = standardized mean difference, (training mean − held-out mean) / pooled *SD*. Positive values indicate higher values in the training partition. SMDs were not computed for the binary characteristics other than the outcome labels. SMDs are

descriptive contrasts between partitions, and intervals were not estimated. No significance tests were conducted. Education is the seven-level code described under Measures. Response scales and item stems are given in the note to Table S1. Complete characteristics and missingness by partition are in Table S8.

Table 2

Pre-Declared Model Selection on the Development Partition

Candidate	Development log loss [95% CI]	Difference from short logistic [95% CI]	Paired bootstrap *SE*	Selection result
Elevated six-month stress				
Intercept only	0.587 [0.559, 0.618]	0.075 [0.044, 0.106]	0.016	Did not displace reference
Short logistic	0.512 [0.474, 0.551]	—	Not applicable	Reference retained
Elastic-net logistic	0.512 [0.476, 0.549]	−0.0001 [−0.009, 0.009]	0.0044	Did not clear margin
Probability average	0.511 [0.475, 0.547]	−0.0014 [−0.010, 0.006]	0.0042	Did not clear margin
Low six-month intention to remain				
Intercept only	0.355 [0.299, 0.421]	0.030 [0.003, 0.060]	0.015	Did not displace reference
Short logistic	0.325 [0.269, 0.387]	—	Not applicable	Reference retained
Elastic-net logistic	0.326 [0.270, 0.387]	0.0009 [−0.003, 0.005]	0.0022	Did not clear margin
Probability average	0.327 [0.271, 0.388]	0.0019 [−0.005, 0.009]	0.0036	Did not clear margin

Note. Models were fit on the training partition and scored on the development partition in the single selection analysis. No model was refit. For stress, $n = 474$, with 127 events. For low intention to remain, $n = 438$, with 50 events. Intervals are 95% percentile intervals from 2,000 bootstrap resamples of development participants (seeds are listed in Supplement S3). The log-loss intervals and the paired differences come from the same bootstrap. Difference from short logistic = candidate log loss minus short-logistic log loss on the same resampled participants. Negative values favor the candidate. Differences were computed on unrounded values and can differ by 0.001 from the difference of the rounded entries. A challenger had to lower log loss by more than one paired bootstrap *SE* to replace the reference. Complete development performance is reported in Table S2.

Table 3

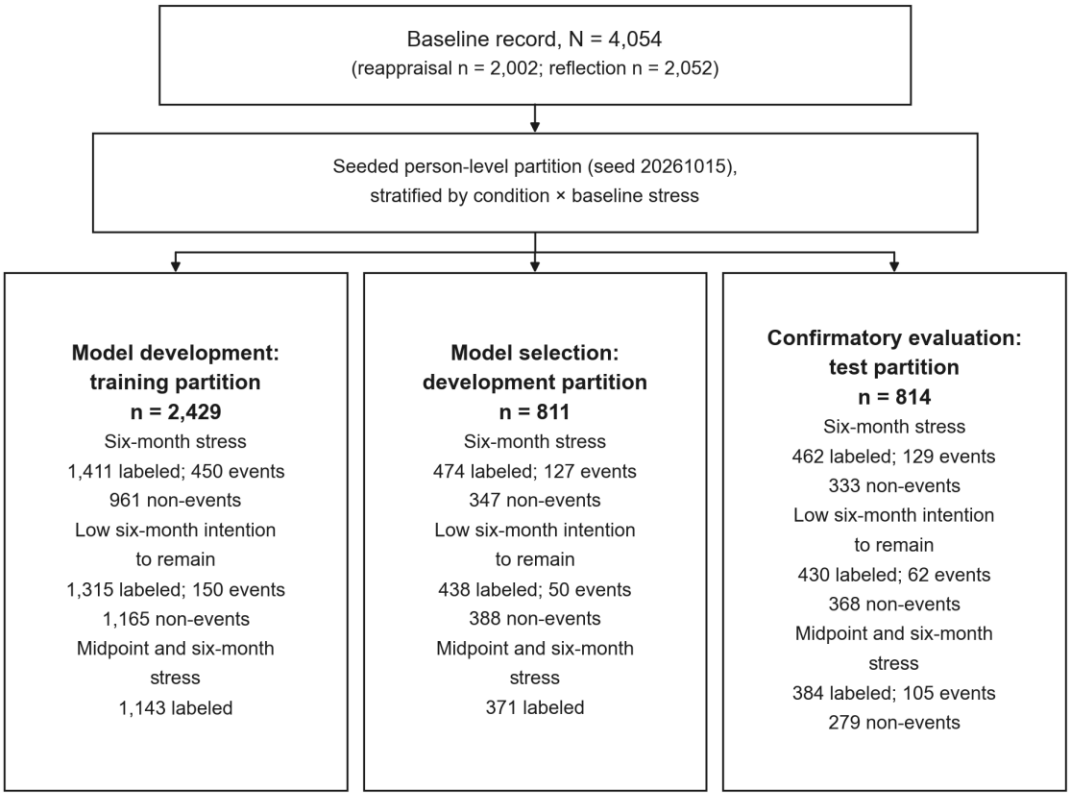
Test-Partition Performance of the Confirmatory Models

Model	Log loss [95% CI]	Brier score [95% CI]	AUROC [95% CI]	AUPRC [95% CI]
H1: Elevated six-month stress (*n* = 462; 129 events)				
Intercept only	0.596 [0.566, 0.627]	.203 [.189, .218]	—	.28 [.24, .32]
Baseline logistic (final)	0.532 [0.494, 0.574]	.177 [.161, .195]	.72 [.67, .77]	.52 [.43, .61]
H2: Low six-month intention to remain (*n* = 430; 62 events)				
Intercept only	0.417 [0.350, 0.488]	.124 [.099, .151]	—	.14 [.11, .18]
Baseline logistic (final)	0.386 [0.323, 0.455]	.118 [.095, .142]	.70 [.63, .77]	.26 [.19, .37]
H3: Elevated six-month stress (*n* = 462; 129 events)				
Baseline elastic net	0.534 [0.496, 0.574]	.177 [.161, .194]	.72 [.67, .77]	.52 [.44, .61]
Session-augmented elastic net	0.532 [0.493, 0.571]	.177 [.160, .193]	.72 [.67, .77]	.53 [.44, .61]
H4: Elevated six-month stress, midpoint subsample (*n* = 384; 105 events)				
Session-augmented elastic net, midpoint-complete refit	0.536 [0.490, 0.583]	.178 [.158, .198]	.71 [.65, .76]	.51 [.42, .61]
Midpoint-augmented elastic net	0.536 [0.485, 0.590]	.177 [.156, .199]	.72 [.66, .77]	.52 [.43, .62]

Note. Models were fit on the training partition and scored in the single confirmatory analysis on the held-out test partition. No model was fit, tuned, updated, or recalibrated on test participants. Intervals are 95% percentile intervals from 2,000 bootstrap resamples of test participants. H1–H4 = Hypotheses 1–4. AUROC = area under the receiver operating characteristic curve. AUPRC = area under the precision–recall curve. AUROC is undefined for an intercept-only model, and its AUPRC equals the event proportion. Paired differences in the text were computed on unrounded values and can differ by 0.001 from the difference of the rounded entries here. Paired log-loss comparisons are shown in Figure 2. Calibration intercepts and slopes are shown in Figure 3.

Figure 1

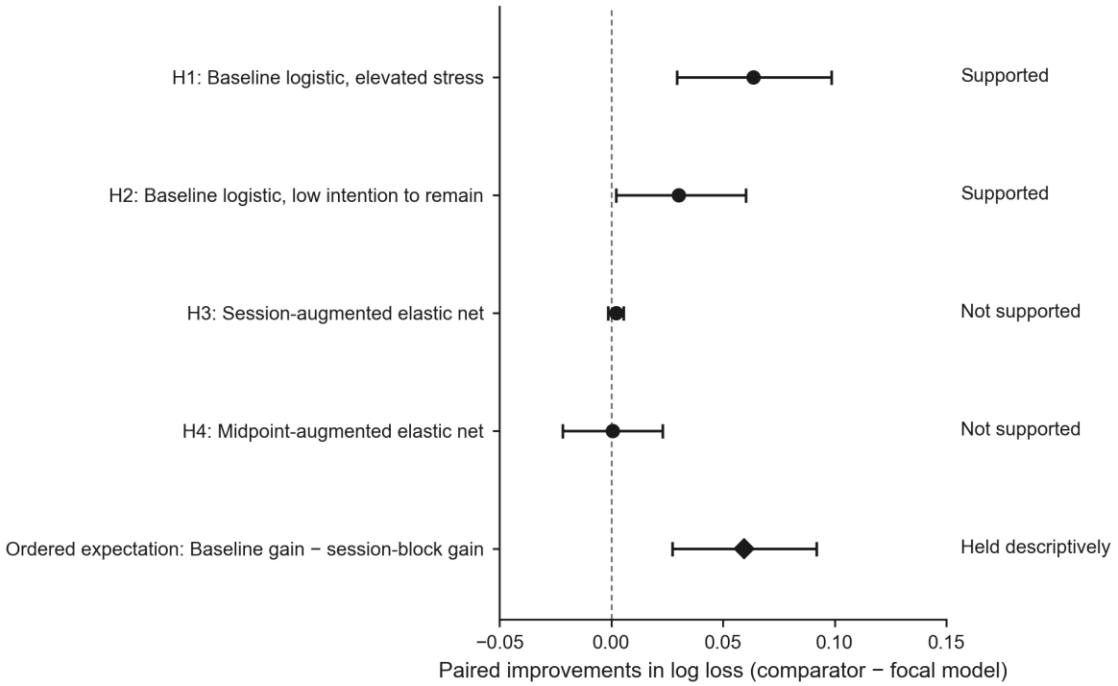
Participant Flow Through Model Development, Selection, and Held-Out Evaluation



Note. Participants were assigned once by person before model fitting, using a seeded 60/20/20 split stratified by assigned condition and baseline stress. Labeled counts are participants with a numeric value on the named six-month outcome. Events are the subset meeting the outcome cut, and the remainder of each labeled count are non-events. Midpoint event counts are shown for the test partition, where they enter Hypothesis 4. The midpoint count requires numeric stress at both midpoint and six months. Median follow-up was 184 days (range 106–248; Wang et al., 2025). The training partition was used for fitting and tuning, the development partition for one comparison of four pre-declared candidates, and the test partition for one pre-specified evaluation.

Figure 2

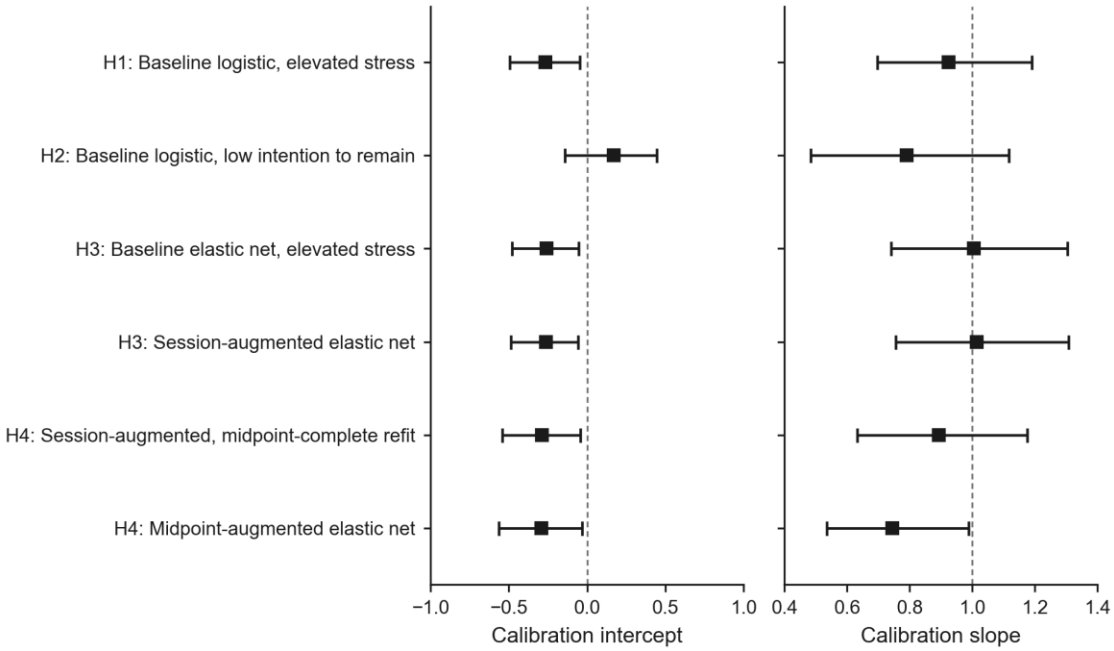
Paired Improvements in Test Log Loss for the Confirmatory Comparisons



Note. Points are paired improvements in test log loss (comparator minus focal model) and lines are bootstrap 95% percentile CIs from 2,000 resamples of the same test participants for both models. Positive values favor the focal model named on the row. The dashed line marks no difference. H1–H4 = Hypotheses 1–4. H1 and H2 compare the final baseline logistic model with the training base-rate model. H3 compares the session-augmented elastic net with its baseline elastic-net comparator. H4 compares the midpoint-augmented specification with its midpoint-complete comparator. The fifth row is the ordered expectation (the baseline elastic net's improvement over the base rate minus the further improvement from the session measures), evaluated descriptively from one joint resample. The pre-declared support criterion also required the relevant point comparisons described in the text.

Figure 3

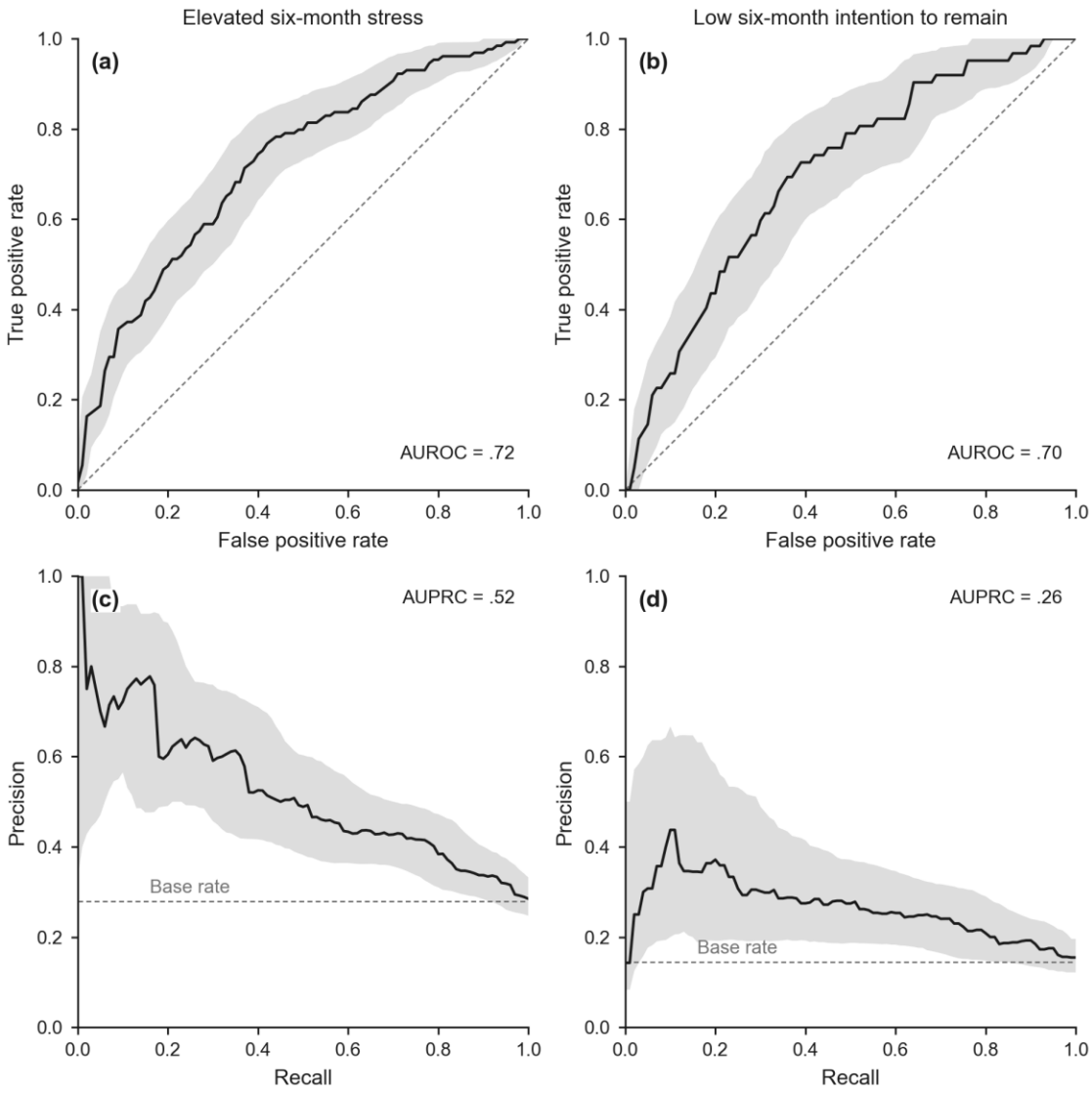
Calibration Intercepts and Slopes of the Confirmatory Models on the Test Partition



Note. Points are calibration intercepts (left) and slopes (right) from logistic recalibration of the outcome on the model logit. Lines are bootstrap 95% percentile CIs from 2,000 resamples. H1–H4 = Hypotheses 1–4. Ideal values are 0 for the intercept and 1 for the slope. A negative intercept indicates predictions that are too high on average, and a slope below 1 indicates predictions that are too extreme. Intercepts and slopes summarize weak calibration. Grouped observed-versus-predicted curves for the same six models are in Figure S4. Development-partition intercepts and slopes for the final model are in Table S2.

Figure 4

Test-Partition Receiver Operating Characteristic and Precision–Recall Curves for the Final Model, With Pointwise Bootstrap Bands



Note. Test-partition curves are for the final 11-predictor logistic model, fit on the training partition and scored once. Elevated six-month stress is shown in panels a and c (n = 462; 129 events). Low six-month intention to remain is shown in panels b and d (n = 430; 62 events). Curves are 101-point grids. They give the true-positive rate at fixed false-positive rates and precision as the scikit-learn step function at fixed recall. Shaded bands are pointwise 95%

percentile intervals from 2,000 bootstrap resamples of test participants (seeds are listed in Supplement S3). Every resample contained both outcome classes. Dashed diagonals mark chance discrimination. Dashed horizontal lines mark the test event rate. The areas under the curves equal the AUROC and AUPRC values in Table 3. No model was refit. That analysis was specified in docs/POST_TEST_ROC_SPEC.md, reviewed under the project's gate procedure, and run once. It returned these gridded aggregates only and no per-participant values.

Supplemental Material for

Stress and Turnover in Helping Professions: A Randomized Controlled Longitudinal Study of Head Start Employees

Matthew J. Monnot

Vocentive

Author Note

S1. Training-Partition Descriptive Statistics and Correlations

Table S1 reports training-partition means, standard deviations, missingness, Spearman–Brown reliabilities, and correlations.

Table S1

Training-Partition Descriptive Statistics, Missingness, Reliabilities, and Correlations

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Baseline stress	2.36	0.95	—												
2. Baseline burnout	2.13	0.99	.65	—											
3. Anxiety	2.02	0.84	.67	.53	(.77)										
4. Depressive symptoms	1.63	0.74	.51	.52	.58	(.71)									
5. Job satisfaction	5.07	1.46	−.41	−.52	−.35	−.32	(.79)								
6. Reappraisal beliefs	5.82	1.16	−.04	−.07	−.06	−.13	.19	—							
7. Reappraisal intentions	5.48	1.13	−.05	−.05	−.03	−.09	.15	.57	—						
8. Hope	5.29	1.40	−.44	−.42	−.46	−.45	.49	.34	.32	—					
9. Age	42.92	12.46	−.14	−.10	−.16	−.11	.08	.03	−.01	.11	—				
10. Female	.95	0.22	.00	.02	.00	−.03	.03	.03	.06	.02	.00	—			
11. Education	5.03	1.51	.10	.07	.04	−.01	−.07	.13	.06	.00	.14	−.01	—		
12. Six-month stress	2.31	0.88	.42	.33	.36	.30	−.26	−.09	−.09	−.33	−.17	−.01	.06	—	
13. Intention to remain	86.18	23.62	−.16	−.20	−.12	−.14	.31	.13	.09	.19	.13	.03	−.09	−.23	—

Note. Training partition only (n = 2,429; pairwise n = 1,301–2,429). Age n = 2,402, education n = 2,378, and female n = 2,423. Other baseline means use n = 2,429. Six-month stress n = 1,411 and intention to remain n = 1,315. Response ranges: stress, burnout, anxiety, depressive symptoms, and six-month stress 1–4; job satisfaction, reappraisal beliefs and intentions, and hope 1–7; age in years; female coded 1 = woman, 0 = otherwise, so *M* is a proportion; education as a seven-level code; intention to remain 0–100. Spearman–Brown reliabilities for the three two-item composites, computed in the training partition (n = 2,429), appear in parentheses on the diagonal; single items have no internal-consistency estimate. Correlations are Pearson, pairwise complete. Item stems from the English initial-

session instrument are as follows. Stress, burnout, anxiety, and depressive symptoms were rated on the matrix "Over the last two weeks, how often have you had the following experiences?" with the stems "Feeling stressed," "Feeling burned out," "Feeling nervous, anxious, or on edge" and "Not being able to stop or control worrying" (anxiety), and "Having little interest or pleasure in doing things" and "Feeling down, depressed, or hopeless" (depressive symptoms). Job satisfaction averaged "Right now, I'm feeling satisfied with my job" and the reverse-scored "Right now, I'm feeling negative about my job." Hope was "Right now, I'm feeling hopeful, optimistic, and encouraged." Reappraisal beliefs was "I believe that changing how I think about a situation can change my emotions," and reappraisal intentions was "I plan to change how I think about a situation when I want to change my emotions." Agreement items were rated 1 = *strongly disagree* to 7 = *strongly agree*. Intention to remain was "How likely are you to continue working in your current organization in the next 12 months?" (0 = *not at all*, 100 = *certainly*). Items in the full baseline list but not in this table were "Feeling happy" (frequency matrix), "In the last two weeks, I've tried changing how I think about a situation in order to feel better" (reappraisal behavior, agreement scale), and the workplace resilience slider "In the last two weeks, to what extent have you been able to bounce back after illness or hardship in the workplace?" (0 = *0% of the time*, 100 = *100% of the time*).

S2. Complete Development-Partition Performance of the Pre-Declared Candidates

Table S2 reports complete development-partition performance.

Table S2

Complete Development-Partition Performance of the Pre-Declared Candidates

Candidate	Log loss [95% CI]	Brier score [95% CI]	AUROC [95% CI]	AUPRC [95% CI]	Calibration intercept [95% CI]	Calibration slope [95% CI]	Difference from short logistic [95% CI]
Elevated six-month stress (*n* = 474; 127 events)							
Intercept only	0.587 [0.559, 0.618]	.199 [.185, .213]	—	.27 [.23, .31]	−0.25 [−0.45, −0.05]	—	0.075 [0.044, 0.106]
Short logistic	0.512 [0.474, 0.551]	.168 [.152, .185]	.74 [.68, .78]	.54 [.46, .63]	−0.27 [−0.50, −0.06]	1.03 [0.79, 1.30]	—
Elastic-net logistic	0.512 [0.476, 0.549]	.168 [.152, .184]	.74 [.69, .79]	.53 [.45, .62]	−0.26 [−0.47, −0.05]	1.15 [0.89, 1.45]	−0.0001 [−0.009, 0.009]
Probability average	0.511 [0.475, 0.547]	.167 [.152, .183]	.74 [.69, .79]	.54 [.45, .63]	−0.26 [−0.48, −0.05]	1.17 [0.90, 1.49]	−0.0014 [−0.010, 0.006]
Low six-month intention to remain (*n* = 438; 50 events)							
Intercept only	0.355 [0.299, 0.421]	.101 [.080, .126]	—	.11 [.09, .15]	0.00 [−0.30, 0.28]	—	0.030 [0.003, 0.060]
Short logistic	0.325 [0.269, 0.387]	.094 [.075, .115]	.71 [.63, .79]	.28 [.19, .42]	−0.04 [−0.38, 0.25]	0.87 [0.53, 1.25]	—
Elastic-net logistic	0.326 [0.270, 0.387]	.094 [.075, .116]	.71 [.63, .79]	.25 [.18, .38]	−0.04 [−0.37, 0.26]	0.92 [0.56, 1.33]	0.0009 [−0.003, 0.005]
Probability average	0.327 [0.271, 0.388]	.095 [.075, .116]	.71 [.63, .79]	.27 [.18, .40]	−0.02 [−0.34, 0.27]	1.01 [0.62, 1.46]	0.0019 [−0.005, 0.009]

Note. Development partition, scored once with models fit on the training partition; no model was refit. Intervals are 95% percentile intervals from 2,000 bootstrap resamples. The log-loss intervals and paired differences are the same values as in Table 2. Difference from short logistic = candidate log loss minus short-logistic log loss on the same resampled participants; negative values favor the

candidate. AUROC = area under the receiver operating characteristic curve; AUPRC = area under the precision–recall curve. Calibration slope and AUROC are undefined for the intercept-only model. The retention rule is stated in Table 2.

S3. Model Specifications, Hyperparameter Search Spaces, and Tuning Procedures

Table S3 reports model specifications and tuning procedures.

Table S3

Model Specifications, Hyperparameter Search Spaces, and Tuning Procedures

Model	Predictor set and preprocessing	Hyperparameters	Validation and final rule
Intercept only	Training event proportion	None	Fit on all labeled training rows
Short logistic	11 baseline predictors; fold-wise median/mode imputation, standardization, and one-hot coding	Unpenalized	Refit within each outer fold; final fit on all labeled training rows
Ridge logistic	Full baseline set; same preprocessing	C = 0.0001 to 100 on the 10-point stored grid	Five-fold inner search by log loss
Elastic-net logistic	Full baseline set; same preprocessing	Same C grid; l1_ratio = 0, 0.5, or 1	Five-fold inner search by log loss
Spline ridge logistic	Full baseline set; numeric predictors expanded with four-knot cubic quantile splines and linear extrapolation	Same C grid	Five-fold inner search by log loss
Random forest	Full baseline set; fold-wise imputation and one-hot coding	n_estimators = 500; min_samples_leaf = 5, 20, or 50; max_features = “sqrt” or 0.5	Five-fold stratified inner search by log loss (scoring = neg_log_loss)
Histogram gradient boosting	Full baseline set; fold-wise imputation	learning_rate = 0.03 or 0.1; max_leaf_nodes = 7, 15, or 31; l2_regularization = 0.0 or 1.0; max_iter = 500; early_stopping = True; validation_fraction = 0.1; n_iter_no_change = 10	Five-fold stratified inner search by log loss (scoring = neg_log_loss)
Probability average	Equal average of short logistic, elastic net, forest, and boosting probabilities	Base-model settings selected inside the outer training fold	No second-level parameters
Stacked ensemble	Same four base models	Logistic final estimator; five-fold stacking	Base settings selected before stacking
Session-augmented elastic net	Baseline plus condition and post-session measures	C = 0.01; l1_ratio = 0	Frozen training fit
Midpoint-augmented elastic net	Session-augmented set plus pre-refresher midpoint measures	C = 0.01; l1_ratio = 0	Frozen fit on 1,143 midpoint-complete training rows

Note. All preprocessing and tuning occurred within the training side of each fold. Full per-fold selected hyperparameters and performance records are stored in output/training_ladder_results.json; this table summarizes rather than replaces that machine-readable record. No development or test participant was used for fitting or tuning.

Random seeds

The split was 60/20/20 with seed 20261015, and the block-assignment rule is in analysis/00_inventory_and_split.py.

Table S3b lists each random seed, its use, and the script that sets it.

Table S3b

Random Seeds and the Scripts That Set Them

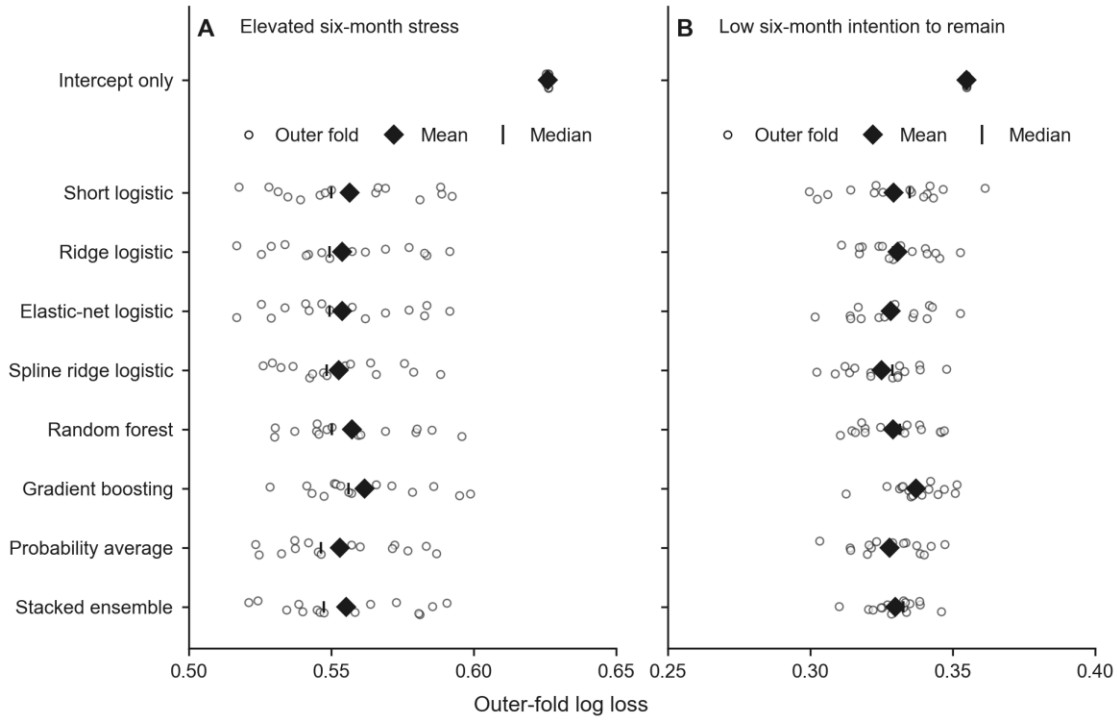
Seed	Use	Script
20261015	Person split, development bootstrap (B = 2,000), training-comparison repeat 1, Figure S2 fold scheme, IPW five-fold split, and instability procedure seed	analysis/00_inventory_and_split.py, 01_fit_train.py, 03_training_ladder.py, 04_fit_specifications.py, 05_dev_test_once.py, 16_attrition_ipw_training.py
20261016	Confirmatory test bootstrap (B = 2,000) as the split seed plus 1, BCa and bootstrap-t, and training-comparison repeat 2	analysis/05_dev_test_once.py, 15_calibration_once.py, 03_training_ladder.py
20261017	Post-test exploratory analysis, test receiver operating characteristic bands, and training-comparison repeat 3	analysis/11_post_test_once.py, 18_roc_once.py, 03_training_ladder.py
20261018	Training bootstrap comparison (B = 100), follow-ups, extra percentile interval, and instability bootstrap	analysis/12_training_bootstrap_ladder.py, 14_training_followups.py, 07_instability_bootstrap.py, 15_calibration_once.py
20261019	Extra percentile interval only	analysis/15_calibration_once.py
20261020	Extra percentile interval only	analysis/15_calibration_once.py

S4. Training-Partition Model Comparison

Repeated stratified 5 × 3 cross-validation compared the nine model classes with tuning repeated inside each outer training fold. Figure S1 displays fold-level log loss; Table S4 reports the complete metric summaries. Both are fold-variation diagnostics. They are not confidence intervals for the fitting procedure, and they do not satisfy TRIPOD+AI item 23a for internal validation.

Figure S1

Outer-Fold Log Loss Across the Training-Partition Model Comparison

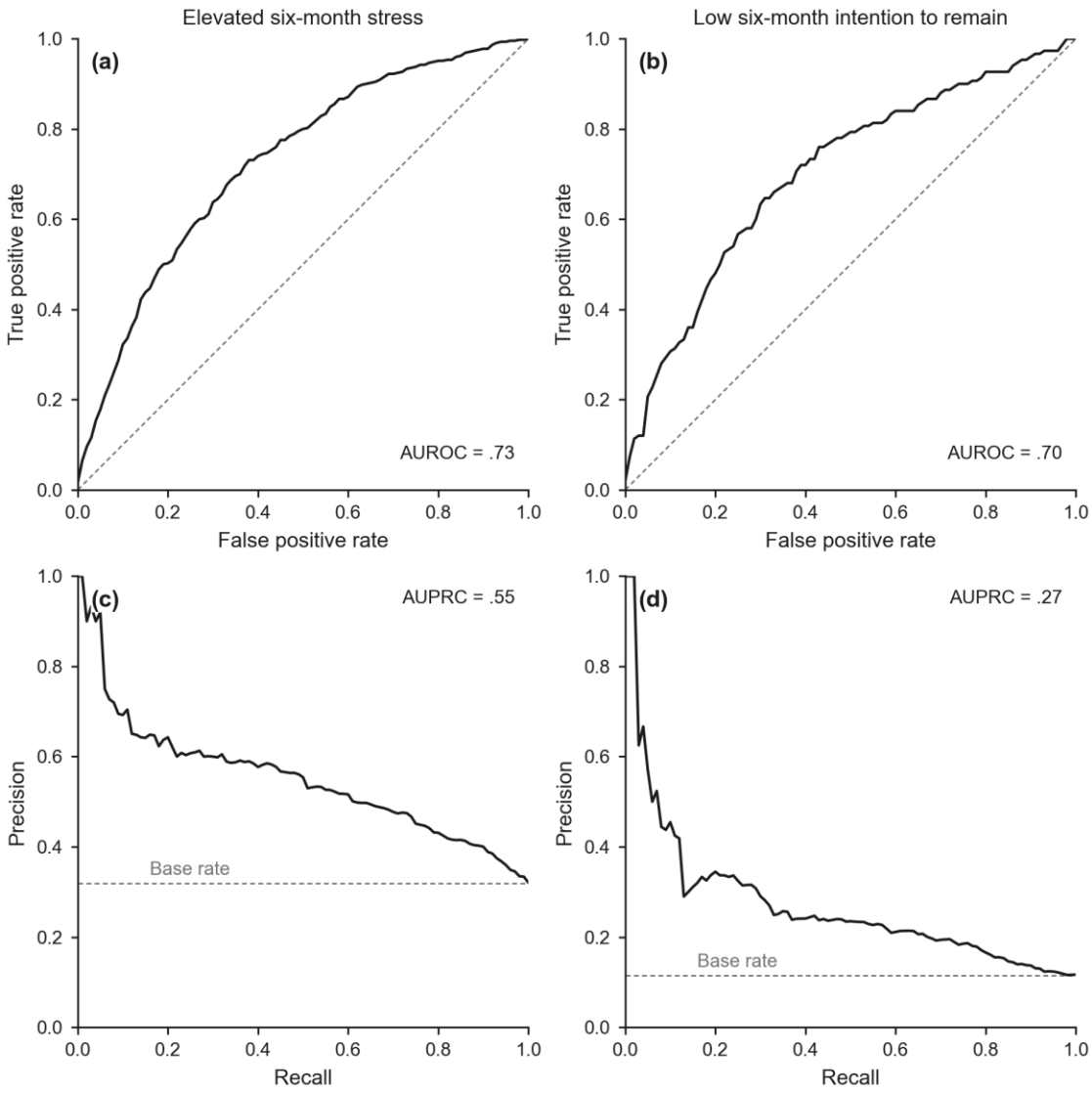


Note. Panel A: elevated six-month stress; Panel B: low six-month intention to remain. Each open circle is log loss in one outer fold from repeated stratified 5-fold cross-validation (three repeats); diamonds are means and short ticks are medians. Hyperparameter tuning was repeated within each outer training fold. Fold results are not independent because training and test sets overlap across repeats, so the display describes fold-to-fold variation and does not treat the 15 values as a confidence interval for one fitted model (Bates et al., 2024). Lower log loss is better.

Figure S2 shows the out-of-fold discrimination curves.

Figure S2

Out-of-Fold Receiver Operating Characteristic and Precision–Recall Curves for the Short Logistic Model on the Training Partition



Note. Out-of-fold curves for the final 11-predictor unpenalized logistic model from stratified five-fold cross-validation on the training partition (seed 20261015, the fold scheme of the pooled internal estimate), for elevated six-month stress (n = 1,411; 450 events; panels a and c) and low six-month intention to remain (n = 1,315; 150 events; panels b and d). Dashed diagonals mark chance discrimination; dashed horizontal lines mark the training event rate. Each curve was

resampled onto 101 evenly spaced false-positive-rate or recall values before display. The development and test partitions were not read.

Table S4

Training-Partition Performance Across the Extended Model Comparison

Model	Log loss *M* (*SD*)	Brier	AUROC	AUPRC	Intercept	Slope	Difference vs. short *M* (*SD*)
Elevated six-month stress							
Intercept only	0.626 (0.000)	.217	—	.32	0.00	—	0.070 (0.024)
Short logistic (reference)	0.556 (0.024)	.188	.73	.55	0.00	0.95	—
Ridge logistic	0.554 (0.023)	.187	.74	.56	0.00	1.10	−0.003 (0.004)
Elastic-net logistic	0.554 (0.023)	.187	.74	.56	0.00	1.10	−0.003 (0.004)
Spline ridge logistic	0.553 (0.019)	.186	.74	.55	0.01	1.11	−0.004 (0.008)
Random forest	0.557 (0.021)	.188	.73	.55	0.00	1.10	0.001 (0.008)
Gradient boosting	0.562 (0.020)	.190	.72	.54	0.01	1.05	0.005 (0.011)
Probability average	0.553 (0.021)	.187	.73	.56	0.01	1.12	−0.003 (0.006)
Stacked ensemble	0.555 (0.023)	.187	.73	.56	0.00	1.06	−0.001 (0.004)
Low six-month intention to remain							
Intercept only	0.355 (0.000)	.101	—	.11	0.00	—	0.026 (0.018)
Short logistic (reference)	0.329 (0.018)	.095	.70	.28	0.00	0.86	—
Ridge logistic	0.331 (0.012)	.095	.70	.28	0.01	1.07	0.001 (0.008)
Elastic-net logistic	0.328 (0.014)	.094	.70	.28	0.00	1.03	−0.001 (0.006)
Spline ridge logistic	0.325 (0.013)	.094	.71	.30	0.00	1.05	−0.004 (0.007)
Random forest	0.329 (0.012)	.095	.71	.28	0.00	1.14	0.000 (0.011)
Gradient boosting	0.337 (0.010)	.097	.68	.23	0.01	1.00	0.008 (0.012)
Probability average	0.328 (0.012)	.095	.71	.28	0.00	1.10	−0.001 (0.007)
Stacked ensemble	0.330 (0.009)	.095	.71	.29	0.00	1.37	0.001 (0.011)

Note. Values summarize 15 outer folds from repeated stratified 5-fold cross-validation with tuning repeated inside each outer training fold. Fold estimates are not independent. Difference = candidate minus short-logistic log loss within the same outer fold; negative values favor the candidate. The pre-declared descriptive tie rule compared the absolute mean paired difference with its fold *SD*. These summaries are internal-validation diagnostics, not performance estimates for the final fitted model, and they are not confidence intervals for the fitting procedure. AUROC = area under the receiver operating characteristic curve. AUPRC = area under the precision–recall curve. Brier = Brier score. Intercept and Slope are the calibration intercept and slope.

Table S4b gives approximate bootstrap intervals for the linear models in the comparison of Table S4 (Riley & Collins, 2023); the note explains why the resample procedure does not yield valid intervals for the tree-based and ensemble models.

Table S4b

Bootstrap Intervals for the Training-Partition Model Comparison

Model	Log loss	Brier score	AUROC
Elevated six-month stress			
Intercept only	0.626 [0.626, 0.626]	.217 [.217, .217]	—
Short logistic (reference)	0.556 [0.532, 0.572]	.188 [.178, .195]	.73 [.70, .76]
Ridge logistic	0.554 [0.527, 0.565]	.187 [.176, .192]	.74 [.72, .77]
Elastic-net logistic	0.554 [0.527, 0.566]	.187 [.176, .192]	.74 [.71, .77]
Spline ridge logistic	0.553 [0.515, 0.558]	.186 [.171, .189]	.74 [.72, .78]
Random forest	0.557 [—]	.188 [—]	.73 [—]
Gradient boosting	0.562 [—]	.190 [—]	.72 [—]
Probability average	0.553 [—]	.187 [—]	.73 [—]
Stacked ensemble	0.555 [—]	.187 [—]	.73 [—]
Low six-month intention to remain			
Intercept only	0.355 [0.355, 0.355]	.101 [.101, .101]	—
Short logistic (reference)	0.329 [0.311, 0.337]	.095 [.089, .097]	.70 [.67, .75]
Ridge logistic	0.331 [0.306, 0.336]	.095 [.088, .097]	.70 [.67, .76]
Elastic-net logistic	0.328 [0.306, 0.337]	.094 [.088, .097]	.70 [.67, .76]
Spline ridge logistic	0.325 [0.297, 0.329]	.094 [.086, .095]	.71 [.69, .78]
Random forest	0.329 [—]	.095 [—]	.71 [—]
Gradient boosting	0.337 [—]	.097 [—]	.68 [—]
Probability average	0.328 [—]	.095 [—]	.71 [—]
Stacked ensemble	0.330 [—]	.095 [—]	.71 [—]

Note. Training partition only. For each outcome, $B = 100$ label-stratified bootstrap resamples of the labeled training rows (seed 20261018); within each resample the full nested cross-validation comparison of analysis/03_training_ladder.py was rerun and pooled out-of-fold metrics were computed (analysis/12_training_bootstrap_ladder.py; output/training_bootstrap_ladder.json). Point estimates are the Table S4 values; brackets are 2.5th and 97.5th percentiles of the resample metrics. The intervals are approximate and shifted toward better performance, because a

bootstrap resample contains duplicated participants, and a participant whose duplicate sits in a fitting fold is partly memorized when scored in a held-out fold. For the four linear models the shift is small, which is why several point estimates sit near the upper bound. For the random forest, gradient boosting, probability average, and stacked ensemble the resample metrics were inflated far beyond the point estimates (resample log losses of roughly 0.2 to 0.47 against point estimates near 0.55 and 0.33), so those intervals do not bound the estimates and are not reported (—). AUROC is undefined for the intercept-only model. AUROC = area under the receiver operating characteristic curve. Fold-to-fold variation for all nine models is in Table S4.

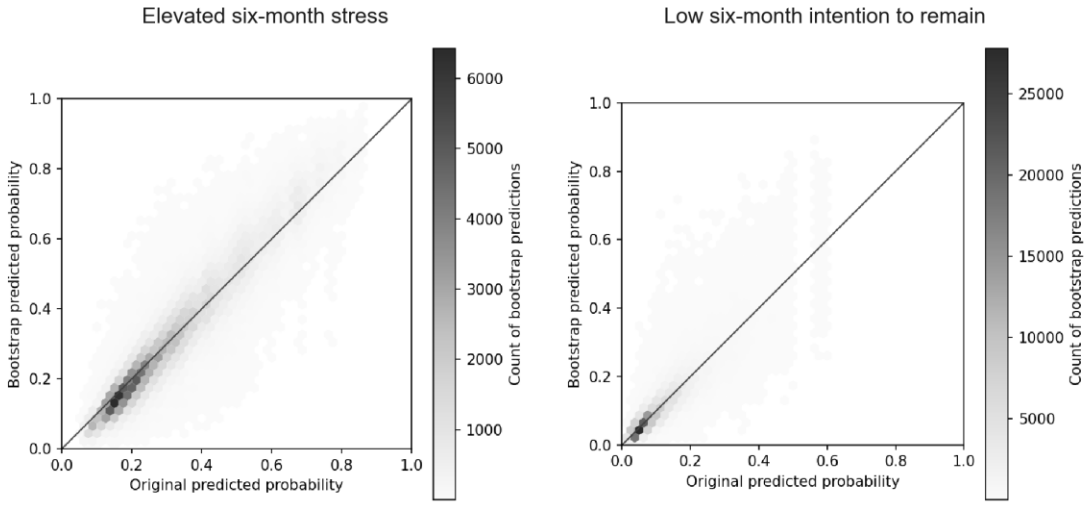
S5. Bootstrap Stability of Individual Predicted Probabilities

Stability of individual predicted probabilities was examined in the training partition only, before any development or test row was read, with the bootstrap procedure of Riley and Collins (2023). Each original model was fit by the baseline elastic-net search, and 200 label-stratified bootstrap samples repeated that fitting procedure.

Figure S3 shows bootstrap stability of predicted probabilities.

Figure S3

Bootstrap Stability of Individual Predicted Probabilities in the Training Partition



Note. Training partition only. Each panel compares bootstrap predicted probabilities with probabilities from the original elastic-net refit; the diagonal is equality. Shading shows the count of bootstrap predictions in each hexagon, with the scale set within each panel. $B = 200$ stratified bootstrap resamples. The stress analysis included $n = 1{,}411$ labeled participants and the intention-to-remain analysis $n = 1{,}315$.

Table S5 reports the bootstrap stability summaries.

Table S5

Bootstrap Stability Summaries for the Training-Partition Elastic-Net Models

Outcome	Selected penalty types across 200 resamples	Selected *C* frequencies	Within-person *SD*, median [25th, 75th, 90th percentile]	Mean absolute prediction difference	Calibration slope *M* (*SD*)
Elevated six-month stress	Ridge 64%; mixed 21%; lasso 15%	0.01: 12%; 0.046: 33.5%; 0.215: 45.5%; 1: 9.0%	0.046 [0.033, 0.059, 0.070]	0.043	0.976 (0.089)
Low six-month intention to remain	Ridge 52.5%; mixed 26%; lasso 21.5%	0.01: 1%; 0.046: 53%; 0.215: 39%; 1: 7%	0.024 [0.016, 0.040, 0.057]	0.025	1.004 (0.163)

Note. Summaries are from 200 stratified bootstrap resamples of the labeled training rows. Within-person *SD* is the standard deviation of bootstrap-model probabilities for the same participant. No development or test predictions are included. Mean absolute differences by original-probability decile are retained in the stored aggregate stability output.

S6. Attrition Within the Training Partition

Within the training partition, participants with versus without a numeric six-month outcome were compared on baseline predictors. Standardized mean differences (SMDs) were the difference in group means divided by the pooled standard deviation; a positive SMD indicates a higher mean among participants who had a numeric six-month item. By assigned arm, 685 of 1,200 reappraisal participants and 726 of 1,229 reflection participants had a numeric six-month stress item.

Table S6 reports the comparison.

Table S6

Standardized Mean Differences at Baseline Between Training Participants With and Without a Numeric Six-Month Outcome

Variable	Stress *SMD*	Intention-to-remain *SMD*
Baseline stress	0.13	0.14
Baseline burnout	0.05	0.08
Anxiety	0.06	0.06
Depressive symptoms	−0.01	−0.02
Job satisfaction	−0.05	−0.06
Reappraisal beliefs	0.22	0.22
Reappraisal intentions	0.04	0.03
Hope	−0.09	−0.10
Age	0.11	0.10
Education	0.33	0.35
Happiness	−0.02	−0.04
Reappraisal behavior	0.16	0.17
Workplace resilience	0.06	0.08

Note. SMD = (mean for participants with a numeric six-month item − mean for participants without that item) / pooled *SD*. The pooled *SD* combines the two sample variances, each with denominator $n - 1$. A positive value means a higher mean among participants who have the item. The stress contrast is $n = 1{,}411$ with numeric stress against $n = 1{,}018$ without it. The intention-to-remain contrast is $n = 1{,}315$ with numeric intention to remain against $n = 1{,}114$ without it. Predictors with missing values, age and education, use the non-missing observations within each group.

S7. Exploratory Analyses on the Test Partition

Table S7 reports the two exploratory analyses pre-specified after the confirmatory analysis (specification available from the author): the baseline short logistic model inside the source authors' exclusion-rule sample, as a refit on included training rows (E1a) and as the frozen confirmatory model (E1b), and elevated six-month burnout as an alternate label (E2). The confirmatory Hypothesis 1 rows are repeated for reference. Both analyses used seed 20261017 and 2,000 bootstrap resamples of the relevant test participants.

The source authors' exclusion rule (E1). Wang et al. (2025) excluded participants flagged by their problem indicator for any session, those scoring 3 or lower on the baseline attention composite, and those who wrote fewer than 15 words on a writing task (at the follow-up sessions, only when the attention slider was also failed). The rule, reproduced from their analysis code, removed 38.0% of the training partition, 40.4% of the development partition, and 38.6% of the test partition (Table S11). The confirmatory analyses kept these participants because the prediction question concerns the staff a program would have data on. Within the exclusion-rule test sample for the stress label (n = 303, 75 events), the baseline logistic model refit on the included training rows had a log loss of 0.492 (95% CI [0.447, 0.540]) against 0.571 [0.538, 0.609] for an intercept model predicting the base rate of the included training rows, a paired difference of −0.079 [−0.118, −0.038], with AUROC .75 [.69, .81] and calibration slope 1.10 [0.80, 1.43]; the frozen confirmatory model, applied without refitting, scored 0.496 [0.450, 0.545], a paired difference of −0.076 [−0.117, −0.034]. The Hypothesis 1 result holds under either sample definition. Both calibration intercepts were more negative than in the full test sample (−0.43 and −0.47 against −0.27), consistent with a lower event rate among included test participants (24.8%) than among all test participants (27.9%).

Burnout as the label (E2). With elevated six-month burnout (a response of 3 or 4; 112 of 462 test participants, 24.2%) in place of stress, the same 11 baseline predictors predicted the outcome better than the base rate. The short logistic model had a log loss of 0.500 [0.448, 0.553] against 0.554 [0.512, 0.596], a paired difference of −0.055 [−0.091, −0.015], and AUROC .73 [.67, .78]; the full-baseline elastic net scored 0.496 [0.445, 0.546], a paired difference of −0.059 [−0.096, −0.021], and AUROC .73 [.68, .78]. Calibration slopes were below 1 for both models (0.80 [0.58, 1.05] and 0.82 [0.61, 1.07]), so their burnout probabilities were somewhat too extreme.

Table S7

Exploratory Analyses on the Test Partition: Exclusion-Rule Sample and Burnout Label

Model	*n*	Events	Log loss [95% CI]	Brier [95% CI]	AUROC [95% CI]	AUPRC [95% CI]	Cal. intercept [95% CI]	Cal. slope [95% CI]	Paired log-loss difference vs intercept [95% CI]
Confirmatory H1 reference									
Intercept only	462	129	0.596 [0.566, 0.627]	.203 [.189, .218]	—	.28 [.24, .32]	−0.19 [−0.39, 0.01]	—	—
Short logistic (retained)	462	129	0.532 [0.494, 0.574]	.177 [.161, .195]	.72 [.67, .77]	.52 [.43, .61]	−0.27 [−0.49, −0.05]	0.92 [0.70, 1.19]	−0.064 [−0.099, −0.029]
E1a exclusion sample, refit									
Intercept only	303	75	0.571 [0.538, 0.609]	.191 [.175, .209]	—	.25 [.20, .30]	−0.34 [−0.59, −0.09]	—	—
Short logistic refit	303	75	0.492 [0.447, 0.540]	.161 [.141, .182]	.75 [.69, .81]	.54 [.43, .65]	−0.43 [−0.72, −0.16]	1.10 [0.80, 1.43]	−0.079 [−0.118, −0.038]
E1b exclusion sample, frozen									
Intercept only	303	75	0.571 [0.533, 0.609]	.191 [.173, .209]	—	.25 [.20, .30]	−0.34 [−0.63, −0.09]	—	—
Short logistic frozen	303	75	0.496 [0.450, 0.545]	.162 [.143, .183]	.75 [.68, .81]	.53 [.41, .65]	−0.47 [−0.75, −0.19]	1.05 [0.75, 1.38]	−0.076 [−0.117, −0.034]
E2 elevated six-month burnout									
Intercept only	462	112	0.554 [0.512, 0.596]	.184 [.165, .203]	—	.24 [.20, .28]	−0.06 [−0.29, 0.14]	—	—
Short logistic	462	112	0.500 [0.448, 0.553]	.163 [.143, .184]	.73 [.67, .78]	.45 [.36, .55]	−0.12 [−0.37, 0.11]	0.80 [0.58, 1.05]	−0.055 [−0.091, −0.015]
Elastic-net logistic	462	112	0.496 [0.445, 0.546]	.161 [.141, .181]	.73 [.68, .78]	.48 [.39, .57]	−0.12 [−0.37, 0.11]	0.82 [0.61, 1.07]	−0.059 [−0.096, −0.021]

Note. Exploratory analyses specified after the confirmatory test analysis. E1 evaluates elevated six-month stress under the source study exclusion rule on included test participants (*n* = 303, 75 events); E1a refits the short logistic model on included training rows, E1b applies the frozen confirmatory coefficients. E2 uses elevated six-month burnout (burnout_w2 >= 3) on all test rows with numeric burnout (*n* = 462, 112 events). Confirmatory H1 row copied from test_once.json for context (*n* = 462, 129 events). Intervals are 95%

percentile bootstrap intervals (B = 2,000). Paired log-loss difference = model minus intercept on the same resampled participants; negative values favor the model. AUROC and calibration slope are undefined for intercept-only models (em dash). Paired differences were computed from unrounded values and may differ in the last digit from the difference of the rounded estimates shown. Log loss and Brier rounded to three decimals; AUROC, AUPRC, and calibration parameters to two decimals, following main-text Table 3.

S8. Complete Characteristics and Missingness Across Data Partitions

Table S8 reports characteristics and missingness in each partition.

Table S8

Complete Characteristics and Missingness Across Data Partitions

Characteristic	Training	Development	Test
Baseline stress	*n* = 2,429; *M* = 2.36 (*SD* = 0.95)	*n* = 811; *M* = 2.36 (*SD* = 0.95)	*n* = 814; *M* = 2.36 (*SD* = 0.95)
Baseline burnout	*n* = 2,429; *M* = 2.13 (*SD* = 0.99)	*n* = 811; *M* = 2.14 (*SD* = 1.04)	*n* = 814; *M* = 2.11 (*SD* = 1.00)
Anxiety	*n* = 2,429; *M* = 2.02 (*SD* = 0.84)	*n* = 811; *M* = 1.98 (*SD* = 0.83)	*n* = 814; *M* = 2.02 (*SD* = 0.85)
Depressive symptoms	*n* = 2,429; *M* = 1.63 (*SD* = 0.74)	*n* = 811; *M* = 1.62 (*SD* = 0.70)	*n* = 814; *M* = 1.63 (*SD* = 0.74)
Job satisfaction	*n* = 2,429; *M* = 5.07 (*SD* = 1.46)	*n* = 811; *M* = 5.04 (*SD* = 1.53)	*n* = 814; *M* = 5.00 (*SD* = 1.52)
Reappraisal beliefs	*n* = 2,429; *M* = 5.82 (*SD* = 1.16)	*n* = 811; *M* = 5.82 (*SD* = 1.13)	*n* = 814; *M* = 5.85 (*SD* = 1.08)
Reappraisal intentions	*n* = 2,429; *M* = 5.48 (*SD* = 1.13)	*n* = 811; *M* = 5.49 (*SD* = 1.13)	*n* = 814; *M* = 5.46 (*SD* = 1.14)
Hope	*n* = 2,429; *M* = 5.29 (*SD* = 1.40)	*n* = 811; *M* = 5.27 (*SD* = 1.43)	*n* = 814; *M* = 5.26 (*SD* = 1.44)
Age	*n* = 2,402; *M* = 42.92 (*SD* = 12.46)	*n* = 804; *M* = 43.46 (*SD* = 12.56)	*n* = 803; *M* = 42.89 (*SD* = 12.41)
Education	*n* = 2,378; *M* = 5.03 (*SD* = 1.51)	*n* = 794; *M* = 5.00 (*SD* = 1.54)	*n* = 791; *M* = 5.11 (*SD* = 1.43)
Female	*n* = 2,423; Female 94.8%	*n* = 810; Female 94.0%	*n* = 808; Female 96.2%
Happiness	*n* = 2,429; *M* = 3.13 (*SD* = 0.84)	—	—
Reappraisal behavior	*n* = 2,429; *M* = 5.34 (*SD* = 1.38)	—	—
Workplace resilience	*n* = 2,429; *M* = 77.06 (*SD* = 23.31)	—	—
Educator	*n* = 2,429; Educator 52.6%	—	—
Management or administration	*n* = 2,429; Not management or administration 81.5%	—	—
Affiliation	*n* = 2,428; HS/EHS 96.5%	—	—
Position	*n* = 2,429; Preschool education 31.4%	—	—
Interview language	*n* = 2,429; English 96.0%	—	—
Six-month stress	*n* = 1,411; *M* = 2.31 (*SD* = 0.88)	*n* = 474; —	*n* = 462; —
Intention to remain	*n* = 1,315; *M* = 86.18 (*SD* = 23.62)	*n* = 438; —	*n* = 430; —
Six-month burnout	*n* = 1,411; *M* = 2.06 (*SD* = 0.89)	*n* = 474; —	*n* = 462; —

Note. Variable definitions match the main-text Measures section. Training-partition cells come from the pre-test aggregate in output/table1_descriptives.json. Development- and test-partition cells come from the pre-specified second analysis (post-test) in

output/partition_descriptives.json. Each cell reports valid *n*; numeric variables report *M* (*SD*) and observed minimum–maximum where the source file provides them; categorical variables report the modal category percent. Em dashes mark quantities that the pre-specified post-test analysis did not return (the additional full-set predictors, and *M*, *SD*, and ranges for the six-month outcome items, on the development and test partitions); they cannot be added without a further analysis of the held-out partitions, which the project's procedure does not permit. Counts below five appear as <5 in the stored record when applicable; modal percents use one decimal. Standardized mean differences by partition are in the stored JSON and are not repeated here. Observed ranges were 1–4 for the frequency items, 1–7 for the agreement items, 18–83 (training), 18–81 (development), and 18–74 (test) for age, and 0–100 for intention and resilience.

Table S8b reports race and ethnicity in the training partition.

Table S8b

Race and Ethnicity in the Training Partition

Response option	*n*	%
White	1,213	50.0
Hispanic or Latino/a	588	24.3
Black or African American	473	19.5
American Indian or Alaska Native	40	1.7
Multiracial	35	1.4
Pacific Islander	25	1.0
Middle Eastern or North African	25	1.0
East Asian	20	0.8
Southeastern Asian	17	0.7
South Asian	10	0.4
Other	36	1.5
I prefer not to say	78	3.2
Selected more than one option	125	5.2
No response	5	—

Note. Item wording: "How do you identify in terms of ethnicity and/or race? Select all that apply." Training partition only (*N* = 2,429); percentages use the 2,424 participants who answered

the item and sum to more than 100 because an option counted as selected whenever it was marked, including for participants who marked several. “Multiracial” is a response option; “Selected more than one option” counts participants who marked two or more options. The other-specify write-in was non-missing for 20 participants and is not reproduced. Race and ethnicity were not used as predictors. Produced by analysis/17_training_race_descriptives.py; aggregates in output/training_race_descriptives.json.

S9. Additional Sensitivity Models on the Training Partition

Two models were fit beside the short logistic reference on the training partition only: an approximate Bayesian logistic regression (MAP + Laplace) with fixed weakly informative priors, and an explainable boosting machine. The four development candidates were fixed before this run and were not altered by it. Development and test partitions were not scored.

The labels, folds, and short logistic reference matched the training comparison reported in the main text. Elevated six-month stress was defined as a stress item of 3 or higher on the 1–4 scale. Low intention to remain was defined as an intention item of 50 or lower on the 0–100 scale. Of 2,429 training participants, 1,411 had a numeric stress label and 450 were at or above the cut (31.9%). For intention to remain, 1,315 had a numeric label and 150 were at or below 50 (11.4%). Outer validation was stratified five-fold cross-validation, repeated three times with seeds 20261015, 20261016, and 20261017. The short logistic model was refit on those folds, and its 15 fold log losses matched the training comparison. The two sensitivity models used the 19-predictor baseline set; the short logistic model used the shorter predictor set from that comparison. Probabilities were clipped to the interval from .000001 to .999999 before scoring. There was no class weighting and no threshold search.

The mean in each table is the mean of the 15 outer-fold scores. The paired difference is the candidate's fold log loss minus the short logistic model's fold log loss, so a negative difference is a lower log loss. A model is tied with the short logistic model when the absolute mean of those 15 differences is smaller than their standard deviation. The *t* statistics are paired *t* values on 14 degrees of freedom. They describe the differences; they are not the tie rule. On 15 folds the tie rule is the same statement as $|t| < \sqrt{15} \approx 3.87$.

Approximate Bayesian logistic regression (MAP + Laplace) used median imputation, mode imputation, standardization to mean 0 and *SD* 1, and one-hot encoding, each fit on the outer training side of the fold. One-hot encoding expanded the design to 35 or 36 coefficients, including the intercept, depending on the fold. Each coefficient, including each one-hot column, had an independent Cauchy(0, 2.5) prior. The intercept had a Cauchy(0, 10) prior (Gelman et al., 2008). Gelman et al. place that coefficient prior on inputs scaled to an *SD* of 0.5. This run kept the training comparison's *SD* of 1 and used the stated Cauchy scales on that scale. The implementation was a maximum a posteriori estimate from L-BFGS-B with an analytic gradient. Numeric predictors were standardized to unit *SD* inside each training fold; one-hot indicators for categorical predictors were left as uncentered 0/1 columns. This is a disclosed adaptation of Gelman et al. (2008), whose default scales continuous inputs to *SD* 0.5 and centers binary inputs. The retained mode was the lowest negative log posterior among a start at the origin and two L2 logistic starts (penalty $C = 1$ and penalty $C = 0.1$). Every fold converged, with a gradient norm below 0.00006. A Laplace approximation used the closed-form Hessian of the negative log posterior. That Hessian was checked on every fold against a numerical Hessian routine; the largest relative gap was under 0.000004, and every Hessian was positive definite without a jitter term. Test-fold probabilities are the probit approximation of the posterior predictive mean, the logistic function of mu divided by the square root of 1 + pi × s2 / 8, where mu and s2 are the mean and variance of the linear predictor under the Gaussian approximation. The approximate Bayesian logistic model had no tuning parameters; inner cross-validation was not run for that model.

The explainable boosting machine (Lou et al., 2013) was fit with ExplainableBoostingClassifier from interpret-core 0.7.8 (Nori et al., 2019). Continuous

predictors were median-imputed and categorical predictors were mode-imputed. Categoricals were passed as nominal features, and continuous predictors were left for the model to bin. A level absent from the fold that fit the imputer was coded as that fold's mode. Imputation for the inner search was fit inside each inner training fold, and the selected model was refit with imputation fit on the outer training fold. The inner grid was interactions in {0, 10} and learning rate in {0.01, 0.05}, scored by clipped log loss on a stratified five-fold split of the outer training fold, using the repeat seed. Max bins (1,024), outer bags (14), and early stopping (100 rounds, tolerance 0.00001, ceiling 50,000 rounds) were left at the package defaults.

Table S9 reports the two sensitivity models.

Table S9

Training-Partition Performance of Additional Sensitivity Models

Panel A: Elevated Six-Month Stress

Model	Log loss *M* (*SD*)	Brier *M* (*SD*)	AUROC *M* (*SD*)	AUPRC *M* (*SD*)	Calibration intercept *M* (*SD*)	Calibration slope *M* (*SD*)	Difference *M* (*SD*)
Short logistic	0.556 (0.024)	.188 (.010)	.73 (.03)	.55 (.05)	0.00 (0.06)	0.95 (0.20)	0.000 (0.000)
Approximate Bayesian logistic	0.559 (0.027)	.190 (.010)	.72 (.04)	.55 (.05)	−0.01 (0.06)	0.89 (0.19)	0.003 (0.007)
EBM	0.548 (0.023)	.185 (.009)	.74 (.04)	.56 (.05)	0.01 (0.06)	1.09 (0.23)	−0.009 (0.006)

Note. n = 1,411 training participants with a numeric label; 450 events. Difference = candidate log loss minus short-logistic log loss. Tie if the absolute mean difference is smaller than the *SD* of the 15 differences. The short logistic row is the reference. The Bayesian model had no inner cross-validation. AUROC = area under the receiver operating characteristic curve. AUPRC = area under the precision–recall curve. Brier = Brier score.

On the stress label, the approximate Bayesian logistic model was tied with the short logistic model. Its mean log loss was 0.559, against 0.556 for the short logistic model and 0.554 for the elastic net in the training comparison. On stress that elastic net reduced to ridge in every fold. The mean paired difference was 0.003, and the *SD* of the differences was 0.007 ($t(14)$ = 1.65). The calibration slope was 0.89. The elastic net slope in the training comparison was 1.10. At this sample size the likelihood dominates a Cauchy(0, 2.5) prior on unit-variance inputs, so the posterior mode stays near a lightly shrunk full logistic. The stronger ridge the elastic net selected is a different fit.

The explainable boosting machine had a lower log loss than the short logistic model, and the absolute mean difference was larger than its fold *SD*: mean 0.548, paired difference −0.009 (*SD* 0.006), $t(14) = -5.86$. The inner search chose 0 interactions in 13 of 15 folds (learning rate 0.05 in 11 of those, 0.01 in two) and 10 interactions in two folds, both with learning rate 0.05. In

those two folds the inner log-loss gain over the best additive setting was under 0.001. The outer-fold gain is in the additive boosted fit. Pairwise interactions were not the source of that gain.

Panel B: Low Six-Month Intention to Remain

Model	Log loss *M* (*SD*)	Brier *M* (*SD*)	AUROC *M* (*SD*)	AUPRC *M* (*SD*)	Calibration intercept *M* (*SD*)	Calibration slope *M* (*SD*)	Difference *M* (*SD*)
Short logistic	0.329 (0.018)	.095 (.004)	.70 (.06)	.28 (.06)	0.00 (0.08)	0.86 (0.29)	0.000 (0.000)
Approximate Bayesian logistic	0.336 (0.016)	.097 (.004)	.68 (.05)	.26 (.05)	−0.08 (0.07)	0.75 (0.23)	0.007 (0.007)
EBM	0.334 (0.011)	.096 (.002)	.68 (.05)	.26 (.05)	0.00 (0.06)	1.04 (0.36)	0.005 (0.009)

Note. n = 1,315 training participants with a numeric label; 150 events. Difference and tie rule are as in Panel A. The Bayesian mean difference and its *SD* both round to 0.007; the unrounded values are 0.0071 and 0.0067, so the absolute mean is not smaller than the *SD*. AUROC = area under the receiver operating characteristic curve. AUPRC = area under the precision–recall curve. Brier = Brier score.

On the intention-to-remain label, the approximate Bayesian logistic model had a higher log loss than the short logistic model, by at least one fold *SD*. The mean was 0.336, against 0.329 for the short logistic model and 0.328 for the elastic net. The calibration slope was 0.75 and the intercept was −0.08, so the predicted probabilities were more dispersed than the fold outcomes. The elastic net selected a stronger penalty than this Cauchy prior, and on this label the Cauchy fit had a higher log loss than the short logistic model.

The explainable boosting machine was tied with the short logistic model. The mean log loss was 0.334, and the paired difference was 0.005 (*SD* 0.009), $t(14) = 1.92$. The inner search chose 0 interactions in 10 of 15 folds and 10 interactions in five. Where 10 interactions won, the inner log-loss gain over the best additive setting was between 0.0008 and 0.0017 (0.000813, 0.000998, 0.001056, 0.001161, and 0.001675). That inner-fold gain did not appear as a lower outer-fold log loss. The point estimate, 0.334, is above the elastic net mean of 0.328.

Calibration slopes here describe the predicted probabilities. Predictor coefficients and effects of condition are separate estimands.

Per-fold explainable boosting machine hyperparameters for each label are stored with the sensitivity analysis outputs. Fold-level scores, paired differences, Laplace diagnostics, and the inner grid in each outer fold are in the stored JSON record for this sensitivity run.

S10. Anticipated Test-Set Precision

Standard errors below are planning values computed from the training partition before any development or test participant was scored. They are not standard errors estimated in a validation sample.

The calculations follow Riley et al. (2021). Equation (4) gives the standard error of ln(O/E), and Equation (5) inverts it. The calibration slope uses the logistic recalibration in Equation (2), the Fisher information in Equation (6), and the sample-size rearrangement in Equation (7). The *C* statistic uses Equation (11) as printed. A check of that implementation reproduces the paper's worked example: $C = .70$, event proportion .10, and a target SE of 0.0255 require 1,154 participants. Wald intervals use estimate ± 1.96 × SE. Half-widths are 1.96 times the unrounded standard error.

What was held fixed

Both calculations used the baseline elastic net, refit on the labeled training rows at the penalty stored in the frozen training specification bundle. The penalty was not searched again.

Elevated six-month stress (stress item ≥ 3): ridge, l1_ratio = 0, penalty $C = 0.01$. Training rows with a numeric label: 1,411. Events: 450 (31.9%).

Low intention to remain (intention item ≤ 50): lasso, l1_ratio = 1, penalty $C = 0.2154$. Training rows with a numeric label: 1,315. Events: 150 (11.4%).

The anticipated test size for stress is 462 participants with a numeric label. For intention to remain it is an expectation, not a count: 814 × (1,315/2,429) = 440.7, the anticipated test pool times the training share of rows with a numeric intention-to-remain label.

Assumptions

Planning treated O/E as 1 and the calibration slope as 1 ($\alpha = 0, \beta = 1$), the default Riley et al. (2021, section 3.2.2) use for weak calibration. The event proportion was the training event proportion. The linear predictor was the empirical linear predictor from the frozen training fit, not a normal draw. For stress its mean was −0.86 (*SD* = 0.82); for intention to remain, −2.29 (*SD* = 0.80). The expectations in Equation (6) were the means of the Box 1 (parts D and E) terms over those values. The mean fitted probability matched the training event proportion to three decimals, which is the check that this distribution sits at the right event rate when the slope is 1.

The development *C* (Riley et al.'s term for the apparent *C* on the model-development data, which in this study means the training partition, not the development partition) was the apparent *C* on the same training rows (stress .746, intention to remain .730). That figure is optimistic because the model was fit on those rows. Section 3.3 says to repeat the calculation at a *C* about .05 lower and to keep the larger sample size. Both are reported. The recommended *n* is the larger one.

The paper's target precisions, used for the required sample sizes, were *SE*(ln(O/E)) = 0.051 (section 3.1.1; a 95% interval for O/E of about 0.90 to 1.10, width 0.2, when O/E = 1), *SE*(slope) = 0.051 (section 3.2; width 0.2), and *SE*(*C*) = 0.0255 (section 3.3; width 0.1). Section 3.1.1 also notes that a width of 0.2 is stringent when the event proportion is near .10, and illustrates *SE*(ln(O/E)) = 0.175 (width about 0.7). The primary target remains *SE*(ln(O/E)) = 0.051; the illustration is reported beside it. Net benefit was not calculated. No decision threshold was pre-specified.

Table S10 reports anticipated precision on the test partition.

Table S10

Anticipated Precision of Test-Partition Performance Estimates

Panel A: Elevated Six-Month Stress

At $n = 462$, and at the training event proportion, the expected number of events was about 147.

Quantity	*SE*	95% CI half-width	*n* for the paper's target
ln(O/E)	0.068	0.133	822
Calibration slope	0.140	0.273	3,458
C statistic, apparent *C* = .746	0.024	0.048	424
C statistic, *C* = .696	0.026	0.051	484

The half-width for ln(O/E) is on the log scale. If O/E is 1, the corresponding 95% interval on the ratio scale is 0.875 to 1.143 (width 0.27). If the slope is 1, the 95% interval is 0.73 to 1.27. The *C* target of *SE* = 0.0255 is met at the apparent *C* ($n = 424$) and requires 484 participants at *C* − .05. Section 3.3 therefore sets the *C* sample size at 484.

The binding target was the calibration slope: 3,458 participants, about 1,103 events at the training event proportion. The primary O/E target required 822 participants (about 262 events). The section 3.1.1 illustration with *SE*(ln(O/E)) = 0.175 required 70, which is not the target used here.

Panel B: Low Six-Month Intention to Remain

The anticipated size, 440.7 participants, is an expectation. At the training event proportion the expected number of events in that group was about 50.

Quantity	*SE*	95% CI half-width	*n* for the paper's target
ln(O/E)	0.133	0.260	2,987
Calibration slope	0.196	0.384	6,509
C statistic, apparent *C* = .730	0.038	0.074	955
C statistic, *C* = .680	0.040	0.078	1,074

If O/E is 1, the 95% interval on the ratio scale is 0.77 to 1.30 (width 0.53). If the slope is 1, the 95% interval is 0.62 to 1.38. The recommended *C* sample size is 1,074, the larger of 955

and 1,074. The primary O/E target required 2,987 participants (about 341 events). The calibration slope required 6,509 (about 742 events) and is again the binding target.

At an event proportion of .114, the primary O/E width of 0.2 corresponds to an absolute difference of about .01 between observed and expected risk. That is the setting in which section 3.1.1 says a width of 0.2 can be too stringent. The illustration they give for an event proportion near .10, *SE*(ln(O/E)) = 0.175, required 254 participants here. The primary target reported above remains the width of 0.2.

Scope

The linear predictor is the in-sample predictor from a penalized fit. If the test sample has a narrower case-mix, the slope and *C* calculations understate the sample size that would be required. No held-out partition was scored at the time these planning values were computed.

S11. Exclusion-Rule Sample Sizes

Table S11 counts participants included and excluded under the source authors' pre-specified exclusion rule for each partition and for samples with numeric six-month stress or burnout.

Table S11

Exclusion-Rule Sample Sizes by Partition

Partition	Sample	*n*	Included	Excluded
Training	Full partition	2,429	1,505	924
Training	Numeric six-month stress sample	1,411	946	465
Training	Numeric six-month burnout sample	1,411	946	465
Development	Full partition	811	483	328
Development	Numeric six-month stress sample	474	297	177
Development	Numeric six-month burnout sample	474	297	177
Test	Full partition	814	500	314
Test	Numeric six-month stress sample	462	303	159
Test	Numeric six-month burnout sample	462	303	159

Note. Counts apply the authors' pre-specified exclusion rule (include_overall; see POST_TEST_SPEC.md). Partition-level counts include all assigned participants; stress and burnout rows restrict to participants with a numeric value on the respective six-month outcome.

S12. Frozen Model Coefficients

The coefficients below are those of the frozen training-fit models that scored the development and test partitions (analysis\01_fit_train.py / 04_fit_specifications.py; the one-shot scripts did not refit). Numeric predictors were standardized to the training mean and *SD* of the labeled training rows for that equation before fitting, so their coefficients are per that training *SD*; the categorical predictor gender entered through one-hot encoding as two indicator columns (female; not female) as stored in the fitted object, so the 11 predictors occupy 12 coefficient columns. Table S12b prints the imputation value, mean, and *SD* used to score a new case. Those constants come from the labeled training rows that fit each equation; they are not the pairwise Table S1 means. Scale is reported per row as stored (standardized or raw).

Table S12 reports the frozen short-logistic coefficients.

Table S12

Coefficients of the Retained Short Logistic Model for Both Labels

Predictor	Scale	Feature code	Elevated six-month stress *b*	OR	Low six-month intention to remain *b*	OR
Baseline stress	standardized	num__stress_w0	0.29	1.33	−0.02	0.98
Baseline burnout	standardized	num__burnout_w0	0.21	1.23	0.13	1.14
Anxiety	standardized	num__anxiety_avg_w0	0.17	1.19	−0.16	0.85
Depressive symptoms	standardized	num__depression_avg_w0	0.10	1.11	0.06	1.06
Job satisfaction	standardized	num__sat_job_avg_pre_w0	−0.04	0.96	−0.63	0.53
Reappraisal belief	standardized	num__rea_bel_pre_w0	−0.01	0.99	−0.12	0.89
Reappraisal intention	standardized	num__rea_int_pre_w0	−0.03	0.97	−0.07	0.93
Hope	standardized	num__hope_pre	−0.21	0.81	−0.03	0.97
Age	standardized	num__age_corrected	−0.25	0.78	−0.28	0.76
Education	standardized	num__edu_corrected	−0.03	0.97	0.24	1.27
Gender [female]	raw	cat__gender_female_female	−0.55	0.58	−0.78	0.46
Gender [not female]	raw	cat__gender_female_not_female	0.20	1.22	−0.77	0.46
Intercept	—	—	−0.35	—	−1.55	—

Note. OR = odds ratio per unit change on the stated scale. Unpenalized logistic models fit on the training partition (*C* = infinity; l1_ratio = 0 for both labels). Feature codes match the fitted preprocessing pipeline. Missing gender was imputed as female. To score a new case, impute each numeric predictor at the median in Table S12b, standardize as (value − *M*) / *SD* using that table's stress or intention column, apply the corresponding coefficients, and invert the logit.

Table S12b

Imputation Medians and Standardization Constants for the Retained Short Logistic Models

Predictor	Imputation median	Stress equation *M*	Stress equation *SD*	Intention equation *M*	Intention equation *SD*
Baseline stress	2.00	2.41	0.92	2.42	0.91
Baseline burnout	2.00	2.15	0.98	2.16	0.98
Anxiety	2.00	2.04	0.83	2.04	0.82

Predictor	Imputation median	Stress equation *M*	Stress equation *SD*	Intention equation *M*	Intention equation *SD*
Depressive symptoms	1.50	1.63	0.73	1.62	0.73
Job satisfaction	5.50	5.04	1.47	5.03	1.46
Reappraisal belief	6.00	5.93	1.05	5.94	1.04
Reappraisal intention	6.00	5.50	1.08	5.49	1.08
Hope	6.00	5.24	1.40	5.23	1.39
Age	43.00	43.51	11.76	43.49	11.67
Education	6.00	5.25	1.38	5.27	1.36

Note. Constants are from the labeled training rows that fit each equation (output/model_coefficients.json). They differ from the pairwise training means in Table S1 because unlabeled rows do not enter the scaler.

Table S13 reports the full-baseline elastic-net coefficients.

Table S13

Coefficients of the Full-Baseline Elastic-Net Models

Predictor	Scale	Feature code	Elevated six-month stress *b*	OR	Low six-month intention to remain *b*	OR
Baseline stress	standardized	num__stress_w0	0.22	1.24	0.00	1.00
Baseline burnout	standardized	num__burnout_w0	0.16	1.18	0.07	1.07
Anxiety	standardized	num__anxiety_avg_w0	0.16	1.17	−0.06	0.94
Depressive symptoms	standardized	num__depression_avg_w0	0.09	1.10	0.00	1.00
Job satisfaction	standardized	num__sat_job_avg_pre_w0	−0.06	0.94	−0.61	0.54
Reappraisal belief	standardized	num__rea_bel_pre_w0	−0.03	0.97	−0.11	0.89
Reappraisal intention	standardized	num__rea_int_pre_w0	−0.06	0.95	−0.08	0.93
Hope	standardized	num__hope_pre	−0.15	0.86	−0.00	1.00
Age	standardized	num__age_corrected	−0.17	0.84	−0.22	0.80
Education	standardized	num__edu_corrected	0.00	1.00	0.19	1.21
Happiness	standardized	num__happy_w0	−0.08	0.92	0.00	1.00
Reappraisal behavior	standardized	num__rea_beh_w0	0.10	1.10	0.08	1.08
Workplace resilience	standardized	num__res_wor_pre_1_w0	−0.04	0.96	−0.05	0.96
Gender [female]	raw	cat__gender_female_female	−0.05	0.95	0.00	1.00
Gender [not female]	raw	cat__gender_female_not_female	0.05	1.05	0.00	1.00
Educator [educator]	raw	cat__educator_educator	0.08	1.08	0.00	1.00
Educator [not educator]	raw	cat__educator_not_educator	−0.08	0.92	0.00	1.00
Management or administration [manage/admin]	raw	cat__manage_admin_manage_admin	−0.01	0.99	0.00	1.00
Management or administration [not manage/admin]	raw	cat__manage_admin_not_manage_admin	0.01	1.01	0.00	1.00
Program affiliation [Head Start/EHS]	raw	cat__binary_affiliation_w0_HS/EHS	−0.02	0.98	0.00	1.00
Program affiliation [not Head Start/EHS]	raw	cat__binary_affiliation_w0_not HS/EHS	0.02	1.02	0.00	1.00

Predictor	Scale	Feature code	Elevated six-month stress *b*	OR	Low six-month intention to remain *b*	OR
Position [Family services]	raw	cat__position_exclusive_Family services	−0.05	0.95	0.00	1.00
Position [Health, mental health, and disabilities]	raw	cat__position_exclusive_Health, mental health, and disabilities	−0.08	0.93	0.00	1.00
Position [Home visitor]	raw	cat__position_exclusive_Home visitor	0.03	1.03	0.00	1.00
Position [Infant toddler education]	raw	cat__position_exclusive_Infant toddler education	0.02	1.02	0.04	1.04
Position [Management and administration]	raw	cat__position_exclusive_Management and administration	−0.01	0.99	0.00	1.00
Position [Missing]	raw	cat__position_exclusive_Missing	−0.00	1.00	0.00	1.00
Position [Multiple options]	raw	cat__position_exclusive_Multiple options	−0.02	0.98	0.00	1.00
Position [Nutrition and food service]	raw	cat__position_exclusive_Nutrition and food service	0.00	1.00	0.00	1.00
Position [Other]	raw	cat__position_exclusive_Other	0.04	1.04	0.00	1.00
Position [Parent or guardian]	raw	cat__position_exclusive_Parent or guardian	−0.00	1.00	0.00	1.00
Position [Preschool education]	raw	cat__position_exclusive_Preschool education	0.08	1.08	0.00	1.00
Position [Transportation]	raw	cat__position_exclusive_Transportation	−0.01	0.99	0.00	1.00
Interview language [EN]	raw	cat__UserLanguage_w0_EN	0.04	1.04	0.00	1.00
Interview language [ES]	raw	cat__UserLanguage_w0_ES	−0.04	0.96	0.00	1.00
Intercept	—	—	−0.84	—	−2.29	—

Note. OR = odds ratio per unit change on the stated scale. Elastic-net logistic models fit on the full baseline predictor set in the training partition. The stress column is the Hypothesis 3 comparator. The intention-to-remain column is the full-baseline elastic net scored in the development comparison for Hypothesis 2; it is not a Hypothesis 3 or 4 comparator. Elevated six-month stress: $C = 0.01$; l1_ratio = 0.0. Low six-month intention to remain: $C = 0.2154$; l1_ratio = 1.0. Shared numeric predictors use the Table S12b constants. Additional numeric predictors: happiness, median 3.00, stress-equation $M = 3.13$ ($SD = 0.83$), intention-equation $M = 3.12$ ($SD = 0.83$); reappraisal behavior, median 6.00, 5.43 (1.28) and 5.45 (1.26); workplace resilience, median 84 (stress equation) or 85

(intention equation), $M = 77.66$ ($SD = 22.49$) and 77.88 (22.07). Missing categorical values were imputed as female, not educator, not management or administration, Head Start/EHS affiliation, preschool education, and English. Predictor labels follow the Measures section where named; remaining rows show the stored feature code.

S14. Training-Only Follow-Ups on the Retained Short Logistic Model

After the confirmatory test analysis, four analyses were run on the retained short logistic model using only the training partition (analysis/14_training_followups.py; seed 20261018). The four are apparent performance, learning curves, leave-one-predictor refits, and an out-of-fold reliability table (Austin & Steyerberg, 2017; Pargent et al., 2023; Raschka, 2018; Steyerberg et al., 2010). Development and test rows were not read. Person-level predictions were not stored. Decision-curve analysis was not run, because no threshold was declared. Permutation importance and Shapley values were not computed.

Apparent performance is the resubstitution score of the model fit and scored on the same labeled training rows. Nested cross-validation means are copied from the stored training comparison (15 outer folds). Optimism is nested minus apparent for log loss and Brier score, and apparent minus nested for AUROC and AUPRC, so a positive value means the resubstitution score was more favorable than the nested estimate. Apparent performance is a diagnostic, not a result (Austin & Steyerberg, 2017).

Summary of results. Apparent log loss on the labeled training rows was 0.549 for stress and 0.318 for intention to remain, against nested-cross-validation means of 0.556 and 0.329 (Table S14a). Out-of-fold log loss flattened once the training side of each fold reached about half its labeled size: 0.564 at 50% versus 0.558 at 100% for stress, and 0.331 versus 0.328 for intention (Table S14b). Dropping age raised stress log loss by 0.004; dropping job satisfaction raised intention log loss by 0.012. The other single-predictor drops moved log loss by 0.003 or less (Table S14c). Training out-of-fold reliability tracked observed rates through the middle of the probability range for stress; bins above .60 were sparse, and for intention most out-of-fold probabilities fell below .20 (Table S14d).

Table S14a

Apparent Performance Versus Nested Cross-Validation for the Retained Short Logistic Model

Outcome	Estimate	Log loss	Brier	AUROC	AUPRC
Elevated six-month stress (n = 1,411; 450 events)	Apparent	0.549	.185	.74	.56
	Nested CV *M*	0.556	.188	.73	.55
	Optimism	0.008	.003	.01	.01
Low intention to remain (n = 1,315; 150 events)	Apparent	0.318	.092	.73	.30
	Nested CV *M*	0.329	.095	.70	.28
	Optimism	0.011	.003	.03	.02

Note. Nested CV means are from the 15 outer folds in Table S4. Optimism = nested − apparent for log loss and Brier score; apparent − nested for AUROC and AUPRC.

Learning curves used the same stratified five-fold, three-repeat schedule as the training comparison. Within each outer training side, a stratified subsample of 25%, 50%, 75%, or 100% of that side was drawn, the short logistic model was refit, and the held-out fold was scored (Raschka, 2018). The question is whether further labeled training rows still reduce log loss.

Table S14b

Learning Curves for the Retained Short Logistic Model

Outcome	Training-side fraction	Mean n train	Log loss *M* (*SD*)	Brier *M* (*SD*)	AUROC *M* (*SD*)
Elevated six-month stress	.25	282	0.589 (0.070)	.195 (.009)	.71 (.03)
	.50	564	0.564 (0.016)	.191 (.007)	.72 (.02)
	.75	846	0.560 (0.016)	.189 (.007)	.72 (.02)
	1.00	1,129	0.558 (0.016)	.188 (.007)	.73 (.02)
Low intention to remain	.25	263	0.350 (0.019)	.099 (.005)	.67 (.05)
	.50	526	0.331 (0.017)	.095 (.005)	.70 (.05)
	.75	789	0.330 (0.015)	.095 (.005)	.70 (.03)
	1.00	1,052	0.328 (0.014)	.094 (.004)	.70 (.03)

Note. Means and *SD*s are across 15 folds (three repeats of five-fold cross-validation). The 1.00 row is a re-run of the short logistic model under seed 20261018 and need not match Table S4 to the last digit.

Each of the 11 predictors was dropped in turn. A positive delta means log loss rose when that predictor was removed.

Table S14c

Leave-One-Predictor Log Loss for the Retained Short Logistic Model

Predictor dropped	Stress log loss *M*	Stress Δ	Intention log loss *M*	Intention Δ
None (full short model)	0.558	—	0.328	—
Baseline stress	0.560	0.003	0.326	−0.001
Baseline burnout	0.559	0.001	0.327	−0.001
Anxiety	0.558	0.000	0.326	−0.001
Depressive symptoms	0.557	0.000	0.327	−0.001
Job satisfaction	0.557	−0.001	0.339	0.012
Reappraisal beliefs	0.557	−0.001	0.328	0.000
Reappraisal intentions	0.557	−0.001	0.327	−0.001
Hope	0.559	0.002	0.327	0.000
Age	0.562	0.004	0.329	0.001
Education	0.557	−0.001	0.329	0.001
Female	0.558	0.000	0.327	−0.001

Note. Δ = mean fold log loss after the drop minus mean fold log loss of the full short model on the same 15 folds (seed 20261018). Positive values mean the dropped predictor was carrying information. These are not unique effects and are not odds ratios; the scoring equation remains Table S12.

Out-of-fold predicted probabilities from one five-fold cross-validation run (seed 20261018) were grouped into ten equal-width bins on [0, 1]. Bins with fewer than five out-of-fold cases are masked.

Table S14d

Ten-Bin Out-of-Fold Reliability for the Retained Short Logistic Model

Outcome	Bin	Predicted range	*n*	Mean predicted	Observed rate
Elevated six-month stress	1	.00–.10	80	.09	.09
	2	.10–.20	379	.15	.16
	3	.20–.30	338	.24	.24
	4	.30–.40	216	.35	.38

Outcome	Bin	Predicted range	*n*	Mean predicted	Observed rate
	5	.40–.50	137	.44	.48
	6	.50–.60	113	.54	.58
	7	.60–.70	76	.64	.53
	8	.70–.80	57	.74	.65
	9	.80–.90	15	.84	.80
	10	.90–1.00	<5	—	—
Low intention to remain	1	.00–.10	775	.05	.06
	2	.10–.20	346	.14	.15
	3	.20–.30	124	.24	.23
	4	.30–.40	46	.35	.26
	5	.40–.50	15	.44	.13
	6	.50–.60	6	.55	.67
	7	.60–.70	<5	—	—
	8	.70–.80	<5	—	—
	9	.80–.90	<5	—	—
	10	.90–1.00	<5	—	—

Note. Training-only reliability diagram from one five-fold cross-validation run; it is not a test-partition calibration plot. Intercept and slope on the test partition remain the confirmatory calibration results (Figure 3). Bins with fewer than five out-of-fold cases are masked.

S15. Attrition Sensitivity on the Training Partition

Of 4,054 baseline participants, 2,347 (57.9%) had a numeric six-month stress item and 2,183 (53.8%) had a numeric intention-to-remain item, so nonresponse was 42% for stress and 46% for intention. In response to outside review, a training-only inverse-probability-weighting sensitivity was pre-specified (as a companion to the post-test calibration specification, available from the author) and run with analysis/16_attrition_ipw_training.py. Development and test rows were not read. The analysis asks whether reweighting labeled training rows toward the full baseline cohort, on baseline information, changes the final model's coefficients or its out-of-fold performance. It cannot speak to staff whose nonresponse depends on something baseline measures do not capture.

For each label, a logistic response model was fit on all 2,429 training rows, predicting whether the participant had a numeric six-month value for that label from the 11 short-model predictors plus assigned condition, interview language, program affiliation, and job position, with the training-only imputation and scaling of the confirmatory pipeline. Stabilized weights (marginal response rate divided by the fitted response probability) were computed for labeled rows and truncated at the 1st and 99th percentiles. The final 11-predictor short logistic model was then refit on the labeled rows with and without those case weights, and both fits were scored out of fold on the same stratified five-fold split (seed 20261015), with each metric computed both unweighted (as the confirmatory metrics were) and weighted.

Table S15 reports the attrition-weighting sensitivity.

Table S15

Response Models, Stabilized Weights, and Weighted Versus Unweighted Refits of the Retained Short Logistic Model

Quantity	Elevated six-month stress	Low six-month intention to remain
Training rows / labeled rows (events)	2,429 / 1,411 (450)	2,429 / 1,315 (150)
Response model AUROC, apparent / out-of-fold	.66 / .64	.66 / .65
Largest response-model coefficients per *SD*	reappraisal beliefs 0.29; education 0.28	reappraisal beliefs 0.31; education 0.29
Stabilized weights before truncation: *M* (*SD*), range	1.00 (0.28), 0.60–3.31	1.00 (0.31), 0.60–4.27
Stabilized weights after truncation: *M* (*SD*), range	0.99 (0.26), 0.70–2.28	0.99 (0.28), 0.67–2.27
Out-of-fold log loss, unweighted model / weighted model	0.556 / 0.556	0.327 / 0.327
Out-of-fold Brier, unweighted / weighted	.188 / .188	.094 / .094
Out-of-fold AUROC, unweighted / weighted	.73 / .73	.71 / .71
Out-of-fold calibration slope, unweighted / weighted	0.96 / 0.92	0.90 / 0.88
Paired fold-wise difference in log loss (weighted − unweighted)	0.000	0.000
Largest coefficient shift per *SD* (weighted − unweighted)	hope −0.04	baseline burnout −0.07; hope −0.07

Note. Out-of-fold metrics are means over five folds scored without weights; the same comparison scored with the stabilized weights gave differences of the same size (log loss 0.000 for both labels; AUROC .000 and .003). Response-model AUROC near .65 indicates that baseline information predicts six-month response only weakly, which is why the weights stay close to 1. Coefficients are per training *SD*. No bootstrap intervals were computed; this is a sensitivity, not a confirmatory result.

Reweighting did not change out-of-fold log loss, Brier score, or discrimination for either label to the reported precision, and the largest coefficient shift was 0.07 per *SD*. Under the fitted

response model, weighting on observed baseline information changed out-of-fold log loss by 0.000 at three decimals. No interval was computed, so this is not a test of no difference. The response model’s own misspecification is not checked here, and the result does not establish that the complete-case test metrics are free of attrition bias. Nonresponse that depends on unmeasured factors, including leaving the program, is untestable with these data.

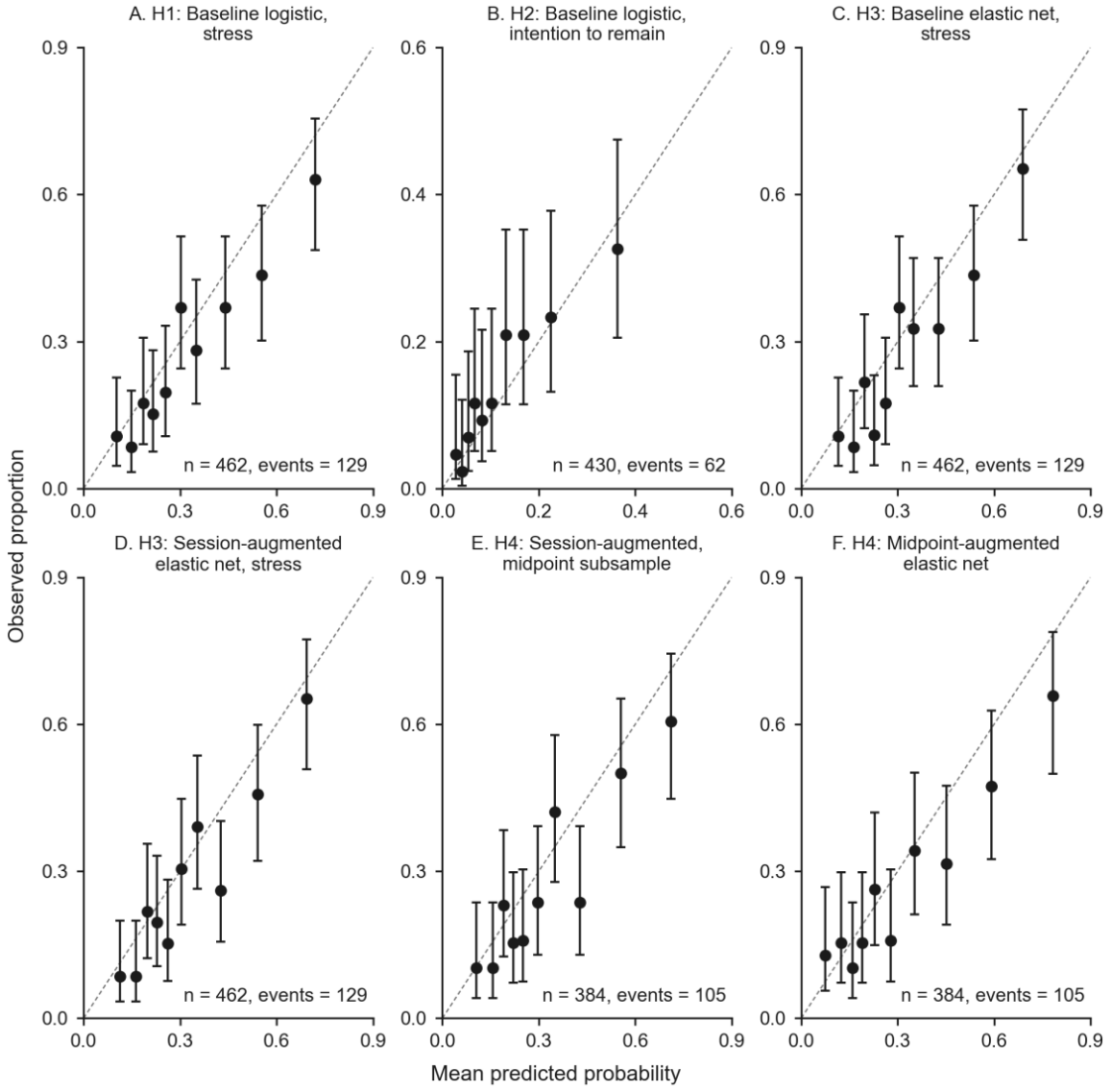
S16. Grouped Calibration on the Test Partition

Grouped calibration for the six confirmatory models was computed in a third, separately pre-specified analysis of the test partition, after the confirmatory analysis and the post-test descriptive analysis. The specification for that analysis is available from the author. Only bin-level aggregates were returned from the held-out analysis: participants, events, mean predicted probability, observed proportion, and a Wilson 95% interval (Wilson, 1927; Brown et al., 2001) for each of 10 equal-width probability bins and 10 quantile deciles. Equal-width bins with fewer than five participants are masked; where exactly one bin is masked, the smallest remaining count on each exact margin is suppressed so the masked cell cannot be recovered by subtraction. Quantile deciles are unmasked. No model was refit or recalibrated. Figure 4 of the main text gives the test-partition discrimination curves from the fourth pre-specified analysis.

Figure S4 shows quantile-decile calibration on the test partition.

Figure S4

Quantile-Decile Calibration of the Six Confirmatory Models on the Test Partition



Note. Panel A: H1 baseline logistic, stress; Panel B: H2 baseline logistic, intention to remain; Panel C: H3 baseline elastic net, stress; Panel D: H3 session-augmented elastic net, stress; Panel E: H4 session-augmented, midpoint subsample; Panel F: H4 midpoint-augmented elastic net. H1–H4 = Hypotheses 1–4. Points are observed proportions by decile of predicted probability, with Wilson 95% intervals; the dashed line is equality. Deciles contain 46–47 participants for n = 462 blocks, 43 for n = 430, and 38–39 for n = 384. Exact values are in Table S16b.

Table S16a reports equal-width calibration bins.

Table S16a

Equal-Width Calibration Bins for the Six Confirmatory Models on the Test Partition

Model	Bin (predicted range)	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
Baseline logistic (H1, stress)	.00–.10	18	1	.08	.06	[.01, .26]
Baseline logistic (H1, stress)	.10–.20	124	17	.16	.14	[.09, .21]
Baseline logistic (H1, stress)	.20–.30	109	21	.24	.19	[.13, .28]
Baseline logistic (H1, stress)	.30–.40	80	26	.34	.33	[.23, .43]
Baseline logistic (H1, stress)	.40–.50	45	16	.45	.36	[.23, .50]
Baseline logistic (H1, stress)	.50–.60	33	14	.55	.42	[.27, .59]
Baseline logistic (H1, stress)	.60–.70	26	13	.65	.50	[.32, .68]
Baseline logistic (H1, stress)	.70–.80	22	17	.74	.77	[.57, .90]
Baseline logistic (H1, stress)	.80–.90	<5	<5	—	—	—
Baseline logistic (H1, stress)	.90–1.00	<5	<5	—	—	—
Baseline logistic (H2, intention to remain)	.00–.10	229	16	.06	.07	[.04, .11]
Baseline logistic (H2, intention to remain)	.10–.20	124	24	.14	.19	[.13, .27]
Baseline logistic (H2, intention to remain)	.20–.30	43	10	.24	.23	[.13, .38]
Baseline logistic (H2, intention to remain)	.30–.40	23	9	.34	.39	[.22, .59]
Baseline logistic (H2, intention to remain)	.40–.50	9	3	.44	.33	[.12, .65]
Baseline logistic (H2, intention to remain)	.50–.60	<5	<5	—	—	—
Baseline logistic (H2, intention to remain)	.60–.70	<5	<5	—	—	—
Baseline logistic (H2, intention to remain)	.70–.80	<5	<5	—	—	—
Baseline logistic (H2, intention to remain)	.80–.90	<5	<5	—	—	—
Baseline logistic (H2, intention to remain)	.90–1.00	<5	<5	—	—	—
Baseline elastic net (H3, stress)	.00–.10	10	1	.09	.10	[.02, .40]
Baseline elastic net (H3, stress)	.10–.20	112	13	.16	.12	[.07, .19]
Baseline elastic net (H3, stress)	.20–.30	126	25	.25	.20	[.14, .28]
Baseline elastic net (H3,	.30–.40	90	31	.34	.34	[.25, .45]

Model	Bin (predicted range)	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
stress)						
Baseline elastic net (H3, stress)	.40–.50	39	12	.45	.31	[.19, .46]
Baseline elastic net (H3, stress)	.50–.60	39	17	.55	.44	[.29, .59]
Baseline elastic net (H3, stress)	.60–.70	30	18	.66	.60	[.42, .75]
Baseline elastic net (H3, stress)	.70–.80	15	11	.74	.73	[.48, .89]
Baseline elastic net (H3, stress)	.80–.90	<5	<5	—	—	—
Baseline elastic net (H3, stress)	.90–1.00	<5	<5	—	—	—
Session-augmented elastic net (H3, stress)	.00–.10	7	1	.08	.14	[.03, .51]
Session-augmented elastic net (H3, stress)	.10–.20	109	13	.15	.12	[.07, .19]
Session-augmented elastic net (H3, stress)	.20–.30	134	25	.24	.19	[.13, .26]
Session-augmented elastic net (H3, stress)	.30–.40	83	29	.34	.35	[.26, .46]
Session-augmented elastic net (H3, stress)	.40–.50	47	15	.45	.32	[.20, .46]
Session-augmented elastic net (H3, stress)	.50–.60	34	15	.55	.44	[.29, .61]
Session-augmented elastic net (H3, stress)	.60–.70	28	16	.64	.57	[.39, .73]
Session-augmented elastic net (H3, stress)	.70–.80	19	14	.75	.74	[.51, .88]
Session-augmented elastic net (H3, stress)	.80–.90	<5	<5	—	—	—
Session-augmented elastic net (H3, stress)	.90–1.00	<5	<5	—	—	—
Session-augmented elastic net (H4, midpoint subsample)	.00–.10	10	1	.08	.10	[.02, .40]
Session-augmented elastic net (H4, midpoint subsample)	.10–.20	99	14	.15	.14	[.09, .22]
Session-augmented elastic net (H4, midpoint subsample)	.20–.30	110	21	.25	.19	[.13, .27]
Session-augmented elastic net (H4, midpoint subsample)	.30–.40	61	21	.35	.34	[.24, .47]
Session-augmented elastic net (H4, midpoint subsample)	.40–.50	31	6	.45	.19	[.09, .36]
Session-augmented elastic net	.50–.60	28	13	.55	.46	[.30, .64]

Model	Bin (predicted range)	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
(H4, midpoint subsample)						
Session-augmented elastic net (H4, midpoint subsample)	.60–.70	25	14	.65	.56	[.37, .73]
Session-augmented elastic net (H4, midpoint subsample)	.70–.80	19	14	.74	.74	[.51, .88]
Session-augmented elastic net (H4, midpoint subsample)	.80–.90	<5	<5	—	—	—
Session-augmented elastic net (H4, midpoint subsample)	.90–1.00	<5	<5	—	—	—
Midpoint-augmented elastic net (H4)	.00–.10	38	suppressed	—	—	—
Midpoint-augmented elastic net (H4)	.10–.20	110	16	.15	.15	[.09, .22]
Midpoint-augmented elastic net (H4)	.20–.30	77	15	.24	.19	[.12, .30]
Midpoint-augmented elastic net (H4)	.30–.40	48	16	.35	.33	[.22, .47]
Midpoint-augmented elastic net (H4)	.40–.50	30	10	.45	.33	[.19, .51]
Midpoint-augmented elastic net (H4)	.50–.60	26	9	.55	.35	[.19, .54]
Midpoint-augmented elastic net (H4)	.60–.70	22	13	.64	.59	[.39, .77]
Midpoint-augmented elastic net (H4)	.70–.80	17	11	.75	.65	[.41, .83]
Midpoint-augmented elastic net (H4)	.80–.90	suppressed	8	—	—	—
Midpoint-augmented elastic net (H4)	.90–1.00	<5	<5	—	—	—

Note. Bins are tenths of the probability scale. Cells with fewer than five participants are masked (<5) and, where exactly one bin in a block is masked, the smallest remaining count on each exact margin is suppressed. Block totals: $n = 462$ (129 events) for the stress models, $n = 430$ (62 events) for intention to remain, $n = 384$ (105 events) for the midpoint subsample.

Table S16b

Quantile-Decile Calibration for the Six Confirmatory Models on the Test Partition

Model	Decile	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
Baseline logistic (H1, stress)	1	47	5	.10	.11	[.05, .23]

Model	Decile	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
Baseline logistic (H1, stress)	2	47	4	.15	.09	[.03, .20]
Baseline logistic (H1, stress)	3	46	8	.19	.17	[.09, .31]
Baseline logistic (H1, stress)	4	46	7	.22	.15	[.08, .28]
Baseline logistic (H1, stress)	5	46	9	.25	.20	[.11, .33]
Baseline logistic (H1, stress)	6	46	17	.30	.37	[.25, .51]
Baseline logistic (H1, stress)	7	46	13	.35	.28	[.17, .43]
Baseline logistic (H1, stress)	8	46	17	.44	.37	[.25, .51]
Baseline logistic (H1, stress)	9	46	20	.55	.43	[.30, .58]
Baseline logistic (H1, stress)	10	46	29	.72	.63	[.49, .75]
Baseline logistic (H2, intention to remain)	1	43	2	.03	.05	[.01, .15]
Baseline logistic (H2, intention to remain)	2	43	1	.04	.02	[.00, .12]
Baseline logistic (H2, intention to remain)	3	43	3	.05	.07	[.02, .19]
Baseline logistic (H2, intention to remain)	4	43	5	.07	.12	[.05, .24]
Baseline logistic (H2, intention to remain)	5	43	4	.08	.09	[.04, .22]
Baseline logistic (H2, intention to remain)	6	43	5	.10	.12	[.05, .24]
Baseline logistic (H2, intention to remain)	7	43	9	.13	.21	[.11, .35]
Baseline logistic (H2, intention to remain)	8	43	9	.17	.21	[.11, .35]
Baseline logistic (H2, intention to remain)	9	43	10	.23	.23	[.13, .38]
Baseline logistic (H2, intention to remain)	10	43	14	.36	.33	[.20, .47]
Baseline elastic net (H3, stress)	1	47	5	.12	.11	[.05, .23]
Baseline elastic net (H3, stress)	2	47	4	.16	.09	[.03, .20]
Baseline elastic net (H3, stress)	3	46	10	.20	.22	[.12, .36]
Baseline elastic net (H3, stress)	4	46	5	.23	.11	[.05, .23]
Baseline elastic net (H3, stress)	5	46	8	.26	.17	[.09, .31]
Baseline elastic net (H3, stress)	6	46	17	.30	.37	[.25, .51]
Baseline elastic net (H3, stress)	7	46	15	.35	.33	[.21, .47]
Baseline elastic net (H3, stress)	8	46	15	.43	.33	[.21, .47]
Baseline elastic net (H3, stress)	9	46	20	.54	.43	[.30, .58]
Baseline elastic net (H3, stress)	10	46	30	.69	.65	[.51, .77]
Session-augmented elastic net (H3, stress)	1	47	4	.11	.09	[.03, .20]
Session-augmented elastic net (H3, stress)	2	47	4	.16	.09	[.03, .20]

Model	Decile	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
Session-augmented elastic net (H3, stress)	3	46	10	.20	.22	[.12, .36]
Session-augmented elastic net (H3, stress)	4	46	9	.23	.20	[.11, .33]
Session-augmented elastic net (H3, stress)	5	46	7	.26	.15	[.08, .28]
Session-augmented elastic net (H3, stress)	6	46	14	.30	.30	[.19, .45]
Session-augmented elastic net (H3, stress)	7	46	18	.35	.39	[.26, .54]
Session-augmented elastic net (H3, stress)	8	46	12	.43	.26	[.16, .40]
Session-augmented elastic net (H3, stress)	9	46	21	.54	.46	[.32, .60]
Session-augmented elastic net (H3, stress)	10	46	30	.69	.65	[.51, .77]
Session-augmented elastic net (H4, midpoint subsample)	1	39	4	.11	.10	[.04, .24]
Session-augmented elastic net (H4, midpoint subsample)	2	39	4	.16	.10	[.04, .24]
Session-augmented elastic net (H4, midpoint subsample)	3	39	9	.19	.23	[.13, .38]
Session-augmented elastic net (H4, midpoint subsample)	4	39	6	.22	.15	[.07, .30]
Session-augmented elastic net (H4, midpoint subsample)	5	38	6	.25	.16	[.07, .30]
Session-augmented elastic net (H4, midpoint subsample)	6	38	9	.30	.24	[.13, .39]
Session-augmented elastic net (H4, midpoint subsample)	7	38	16	.35	.42	[.28, .58]
Session-augmented elastic net (H4, midpoint subsample)	8	38	9	.43	.24	[.13, .39]
Session-augmented elastic net (H4, midpoint subsample)	9	38	19	.56	.50	[.35, .65]
Session-augmented elastic net (H4, midpoint subsample)	10	38	23	.71	.61	[.45, .74]
Midpoint-augmented elastic net (H4)	1	39	5	.08	.13	[.06, .27]
Midpoint-augmented elastic net (H4)	2	39	6	.12	.15	[.07, .30]
Midpoint-augmented elastic net (H4)	3	39	4	.16	.10	[.04, .24]
Midpoint-augmented elastic net (H4)	4	39	6	.19	.15	[.07, .30]
Midpoint-augmented elastic net (H4)	5	38	10	.23	.26	[.15, .42]
Midpoint-augmented elastic net (H4)	6	38	6	.28	.16	[.07, .30]
Midpoint-augmented elastic net (H4)	7	38	13	.35	.34	[.21, .50]
Midpoint-augmented elastic net (H4)	8	38	12	.45	.32	[.19, .47]

Model	Decile	*n*	Events	Mean predicted	Observed proportion	Wilson 95% CI
Midpoint-augmented elastic net (H4)	9	38	18	.59	.47	[.32, .63]
Midpoint-augmented elastic net (H4)	10	38	25	.78	.66	[.50, .79]

Note. Deciles are formed within each block on the predicted probability; sizes differ by one where n is not divisible by 10. Mean predicted and observed values are rounded from the stored aggregates (output/calibration_deciles.json).

S17. Additional Interval Constructions for the Hypothesis 1 and 2 Paired Differences

The confirmatory criterion for Hypotheses 1 and 2 was fixed on the percentile bootstrap interval with seed 20261016 (B = 2,000). At outside review one reader asked whether the Hypothesis 2 log-loss interval, whose upper bound was −0.002, depended on that construction. The same pre-specified analysis recomputed the paired differences (baseline logistic minus training-base-rate intercept) under three further percentile seeds, a bias-corrected and accelerated interval with jackknife acceleration over test participants (Efron, 1987; DiCiccio & Efron, 1996), and a bootstrap-t interval with an inner B of 200 (DiCiccio & Efron, 1996), each generator started fresh at its stated seed. These are reported as a sensitivity; none can change a confirmatory result.

Table S17 reports the paired differences under other interval constructions.

Table S17

Paired Log-Loss and Brier Differences Under Alternative Interval Constructions

Hypothesis	Metric	Point	Percentile, seed 20261016 (confirmatory)	Percentile, seed 20261018	Percentile, seed 20261019	Percentile, seed 20261020	BCa (seed 20261016)	Bootstrap-t (seed 20261016)
H1	Log loss	−0.064	[−0.099, −0.029]	[−0.100, −0.031]	[−0.098, −0.029]	[−0.097, −0.028]	[−0.097, −0.027]	[−0.097, −0.029]
H1	Brier	−.026	[−.041, −.010]	[−.042, −.012]	[−.041, −.011]	[−.041, −.010]	[−.040, −.010]	[−.041, −.011]
H2	Log loss	−0.030	[−0.060, −0.002]	[−0.059, −0.002]	[−0.057, −0.003]	[−0.059, −0.003]	[−0.060, −0.004]	[−0.060, −0.003]
H2	Brier	−.007	[−.015, .002]	[−.016, .002]	[−.014, .001]	[−.016, .001]	[−.016, .001]	[−.016, .001]

Note. Negative values favor the baseline logistic model. B = 2,000 outer resamples throughout; bootstrap-t uses 200 inner resamples. BCa $z0$ and acceleration are in the stored JSON. The Hypothesis 2 Brier interval includes zero under every construction, as the confirmatory interval did; the Hypothesis 2 log-loss interval excludes zero under every construction.

S18. Baseline Predictors and Six-Month Outcome Rates by Demographic Group (Training Partition)

Table S18 reports baseline predictor means and six-month outcome rates by gender, survey language, and race and ethnicity in the training partition.

Table S18

Baseline Predictors and Six-Month Outcome Rates by Demographic Group in the Training Partition

Panel A. Predictor Means by Group

Predictor	Female	Not female	English	Spanish	White	Hispanic or Latino/a	Black or African American
Baseline stress	2.36 (0.95) [0.00]	2.36 (0.96) [0.00]	2.37 (0.96) [0.29]	2.09 (0.66) [−0.29]	2.51 (0.93) [0.33]	2.30 (0.89) [−0.09]	2.07 (0.96) [−0.39]
Baseline burnout	2.13 (0.99) [0.11]	2.02 (0.96) [−0.11]	2.13 (1.00) [0.09]	2.04 (0.82) [−0.09]	2.15 (1.01) [0.04]	2.15 (0.98) [0.03]	2.05 (0.98) [−0.10]
Anxiety	2.02 (0.83) [−0.02]	2.03 (0.92) [0.02]	2.03 (0.84) [0.31]	1.77 (0.65) [−0.31]	2.11 (0.83) [0.23]	1.98 (0.82) [−0.06]	1.84 (0.84) [−0.27]
Depressive symptoms	1.62 (0.74) [−0.13]	1.72 (0.74) [0.13]	1.63 (0.74) [0.13]	1.54 (0.63) [−0.13]	1.64 (0.74) [0.02]	1.63 (0.71) [0.01]	1.56 (0.76) [−0.12]
Job satisfaction	5.08 (1.46) [0.14]	4.89 (1.49) [−0.14]	5.06 (1.46) [−0.28]	5.47 (1.27) [0.28]	4.97 (1.48) [−0.15]	5.24 (1.43) [0.15]	5.21 (1.43) [0.12]
Reappraisal beliefs	5.83 (1.15) [0.13]	5.68 (1.19) [−0.13]	5.83 (1.13) [0.22]	5.58 (1.60) [−0.22]	5.84 (1.06) [0.03]	5.81 (1.25) [−0.01]	5.83 (1.26) [0.01]
Reappraisal intentions	5.49 (1.12) [0.29]	5.17 (1.30) [−0.29]	5.48 (1.12) [0.11]	5.35 (1.44) [−0.11]	5.40 (1.06) [−0.14]	5.54 (1.15) [0.07]	5.62 (1.22) [0.16]
Hope	5.30 (1.40) [0.09]	5.17 (1.40) [−0.09]	5.27 (1.40) [−0.45]	5.90 (1.37) [0.45]	5.08 (1.39) [−0.30]	5.42 (1.40) [0.12]	5.68 (1.37) [0.34]
Age	42.92 (12.38) [−0.01]	43.10 (13.92) [0.01]	42.82 (12.55) [−0.20]	45.34 (9.84) [0.20]	43.29 (12.61) [0.06]	39.96 (11.51) [−0.32]	44.96 (12.68) [0.20]
Education	5.03 (1.51) [−0.06]	5.11 (1.57) [0.06]	5.03 (1.51) [0.05]	4.96 (1.53) [−0.05]	5.09 (1.54) [0.08]	4.84 (1.54) [−0.17]	4.96 (1.51) [−0.06]
Female	—	—	94.8%	93.8%	96.5%	95.1%	96.0%

Panel B. Outcome Rates and n by Group

Group	*n*	Elevated six-month stress	Low six-month intention to remain
Female	2,296	431 / 1,370 (31.5%)	145 / 1,278 (11.3%)
Not female	127	19 / 41 (46.3%)	5 / 37 (13.5%)
English	2,332	442 / 1,362 (32.5%)	suppressed / 1,271
Spanish	97	8 / 49 (16.3%)	<5 / 44
White	1,213	276 / 821 (33.6%)	86 / 783 (11.0%)
Hispanic or Latino/a	588	92 / 286 (32.2%)	27 / 260 (10.4%)
Black or African American	473	57 / 238 (23.9%)	28 / 215 (13.0%)

Note. Training partition only (N = 2,429). Race and ethnicity groups are not mutually exclusive because the item was select-all. A continuous cell is the mean and *SD*, and the bracketed value is the standardized mean difference (group mean − complement mean) / pooled *SD*. The pooled *SD* combines the two sample variances, each with denominator $n - 1$. Positive bracketed values indicate a higher mean in the named column than in its complement. The complement of a gender or language column is the other reported level, and the complement of a race or ethnicity selection is every training participant who did not select that option. Six participants with no gender response are in neither gender column. Female is the percent among participants with a non-missing gender item, and that row is not filled in the gender columns because those columns are that item. Age and education use the non-missing values in each column. Cells with *n* under 5 are written as <5, and the percent is omitted when the event count is under 5. No significance tests were conducted. Outcome rates are among participants with a numeric six-month label. The English low intention to remain cell is suppressed as the complement of the suppressed Spanish cell because the two counts sum to the printed training marginal.

Supplemental References

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